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Master Thesis

Synthesizing Object Dynamics from Human Motion with a Video Generative Prior

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Contents

Abstract

We address the challenge of synthesizing realistic videos that depict dynamic interactions between humans and objects within a scene. Despite numerous efforts in generating human-object motion directly within 3D environments, a complete solution requires not only motion generation but also precise appearance models for effective rendering. More importantly, creating physically plausible 3D motions is inherently complex, requiring either sophisticated physical simulation tools or extensive training datasets.

In this master thesis, we explore the capabilities of large video foundation models to both act as simulators as well as neural renderers, capturing the complex relationship between human motion and object motion dynamics. Specifically, we propose a novel approach for image-based human-object interaction that incorporates a ControlNet model into Stable Video Diffusion, which we fine-tune on the DexYCB human-object interaction dataset.

Through a series of experiments, we investigate the impact of various model architectures and training strategies on video generation performance. We highlight the importance of careful consideration in the design of control signals, information flow, and training strategies to achieve optimal performance and generalization.

Our experimental results not only demonstrate our model's capability to produce high-fidelity, dynamic human-object interactions, but also affirm that foundation models can indeed function as both effective simulators and renderers in the synthesis of complex interactive scenes.

List of Figures

List of Tables

1 Introduction

Humans effortlessly interact with the world around them, performing actions that seem simple on the surface but actually involve a sophisticated understanding of physics, motion, and the interplay between themselves and their environment. Consider the seemingly mundane act of walking to the kitchen, picking up a cup of coffee from the table, and drinking it just before reading this manuscript. Each of these actions involves a sequence of movements and interactions: walking, grasping the cup, bringing it to your mouth, and tilting your head to take a sip. Now imagine an alternate scenario where you slightly retract the fingers holding the cup, causing it to fall and spill. You probably would have an awful day, but as a true optimist and analyst you might in addition make the observation that interaction with objects requires a very deep understanding and precise orchestration of contact and motion. Seemingly simple tasks *mask* the complex dance of physics, motion, and contact – a dance that has proven difficult to replicate in the digital realm.

Human-object interaction (HOI), a field focused on digitally re-creating such complex interaction scenarios therefore often relies on data-driven approaches, which require highly accurate 3D datasets or physical simulations to train neural networks to generate realistic 3D human and object interactions. The inherent complexity of this task, along with the challenges of capturing 3D data in the first place, makes it difficult to capture the nuances of real-world interactions. Modeling the body and hand movement has complexities in itself, but on top of that, when you add interacting objects you need to replicate how these get affected when moved by a human. Moreover, the type of object movement that will happen is very correlated with the intention of the person, something that is often hard to define computationally. Moreover, achieving a complete solution to HOI goes beyond motion generation for 3D models; it requires precise appearance models. For instance, accurately depicting deformations of pliable objects when they are picked up. Importantly, creating not just realistic-looking motions but also physically plausible 3D motions requires advanced physical simulation tools. These tools ensure that the resulting interactions follow the laws of physics, resulting in a higher degree of realism in the final output. The field of 3D HOI is incredibly compelling due to its potential applications in robotics, virtual reality, and animation. However, the limitations mentioned above pose significant challenges in achieving truly realistic interactions.

Figure 1: Motivation. Empirical evidence that a state-of-the-art video generation model can synthesize HOI from an initial single image. Bounding boxes indicate human-object interaction. Generated using Stable Video Diffusion (SVD) [4], better viewed as videos.

Drawing inspiration from recent breakthroughs in image and video foundation models [4, 5, 13, 52, 77], we are interested in approaching human-object interaction in a different manner. Exploratory and empirical evidence suggests that video generation models possess the ability to generate videos with complex physical interactions [23] and generate plausible human-object interactions from the context of a conditioning frame, see Figure 1. Additionally, advancements in multi-view image generation using video models as a 3D prior, hint at its ability to infer the 3D geometry of objects [67] - a property that would definitely come in handy for video-based HOI generation. Video-based HOI would bypass the need for complete 3D models, physics simulations, and explicit rendering, capitalizing instead on the rich understanding of video foundation models.

1.1 Context

When you think about a field in which human-object interaction can make a huge difference, animation is one. Usually animators would start with the 3D modeling of characters: capturing the nuances of facial features, expressions, and skin texture; simulating the complex movement and interaction of individual hairs; accurately representing the human form, including musculature and skeletal structure; and crafting realistic garments that drape, fold, and react to movement. In addition to character creation, the entire surrounding scene must be meticulously constructed to achieve photo-realistic results. This includes: creating buildings, landscapes and interiors with detailed textures and materials. Once the human models, object models and scene are complete, animators give life to the scene by adjusting joint angles and object positions to create realistic motions. Here, Mocap data or a reference video often serve as a guideline. Lastly, the 3D scene is rendered using techniques such as ray tracing or path tracing to simulate the interplay of light and shadow, creating depth and realism. From start to end, this entire pipeline makes the creation of realistic human-object interactions very time-consuming and costly.

One research direction within the field of Human-Object Interaction (HOI) synthesis, focuses on the complexities of the interaction step within this pipeline. Instead of requiring an animator to manually manipulate every body part of a 3D human and a 3D object over time, the objective is to train a neural network to generate such interactions in a realistic manner. This approach essentially maintains explicit spatial and rendering aspects, while modelling the movements of a 3D human and a 3D object implicitly. Such a neural network would predict the pose, rotation, and translation of both the 3D human and the 3D object over time, ensuring realistic and physically plausible interactions. Training such neural networks necessitates data comprising precise 3D models of humans, objects, and interactions between these models with millimeter-level accuracy. Acquiring such data in real-world scenarios poses significant challenges, leading to a reliance on lab-generated HOI training data, see Figure 3. This data is typically produced using 3D point-tracking setups, which are expensive to establish, require specialized knowledge to operate, and generate data modalities that are time-consuming to pre-process and clean. The consequence of these challenges is that existing 3D captured HOI datasets are relatively small and lack diversity in the human-object interactions they capture.

In an ideal world, we would have access to an extensive amount of highly accurate 3D capture data of humans interacting naturally with 3D in-the-wild scenes. However, such comprehensive data is currently unavailable, and methods aiming to provide it are often not easily scalable. Instead, we have access to a vast amount of video data that inherently captures world dynamics, including complex human-object interactions in natural settings. While abundant, this video data lacks ground truth for 3D human and object poses, making it challenging to utilize effectively.

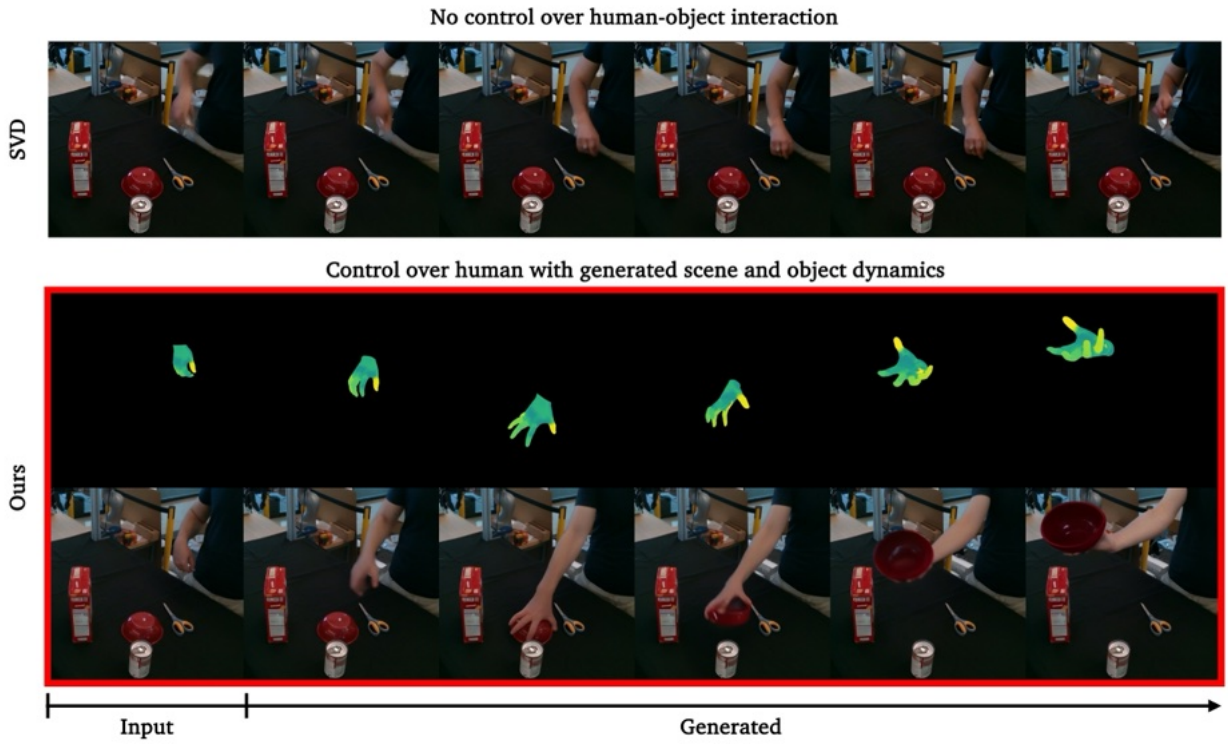


Figure 2: Video-based Human-object Interaction. We train a controllable version of SVD [4] based on ControlNet [77] to generate object dynamics from a human motion control signal. The scene, human motion and object dynamics are fully generated by the diffusion model.

1.2 Goal

Foundational video generation models present a *paradigm shift* in capturing the dynamics of the world. Rather than training a model to perform explicit asset reconstruction, such models are trained end-to-end to capture and represent a data distribution. Powered by extensive training datasets, these models have a profound capacity to learn and reproduce real-world imagery and video at a pixel level, negating the necessity for physics-based models. They not only generate realistic video sequences but also decode complex patterns, structures, and interrelationships within the data. This shift towards end-to-end trained generative models circumvents traditional dependencies on labeled data for training, enabling these systems to learn and infer the underlying physical laws and interactions from the data itself. Compared to traditional frameworks, such as 3D HOI synthesis that only models motion implicitly, video generation models capture a high-dimensional latent space that implicitly models space, time, physics and rendering, all at the same time. Consequently, this learned knowledge can be seen as a very rudimentary form of a *world model* [1, 15, 43]. *Video diffusion models*, in particular, have shown evidence of remarkable 3D understanding [4, 67], and can maintain attribute's invariance over time, thanks to their temporally consistent nature, using only text or a single image as input. Beyond this, we hypothesize that video diffusion models can reason about and understand human-object interaction.

Out-of-the-box, however, most text-to-video (T2V) or image-to-video (I2V) models do not provide a precise control mechanism. For instance, while we can generate a video of a person playing the guitar, we do not have fine-grained control over how exactly they move their hand or fingers. Moreover, since video generation models have an implicit representation of space, time, physics and rendering, we cannot easily gain control over the scene, the movements of humans and objects, like we can in explicit modeling. Therefore, there is a need for a method that can tap into its implicit representation in a way that allows for such precise control. Specifically, we explore the scenario in which we start from a single image of a *scene* with a *human* and an *object*, and generate a video of an *interaction* that is aligned with a given *control signal*. On a practical level, we are interested in using a video generation model as a 'world model', conditioning video generation on an initial frame and a human motion control signal, while entirely relying on the video generation model to simulate the object dynamics, human movement and scene changes. On a high-level, we evaluate the video generation model's ability to perform HOI synthesis, appearance modelling of the object, and implicit rendering of the scene, simultaneously.

1.3 Project

To address the challenge of HOI synthesis in a different way, we propose leveraging a video diffusion model to generate human-object interactions. Specifically, we are interested in starting from a single image of a *scene* with a *human* and an *object*, from which the diffusion model generates a video of an *interaction* that is aligned with a *control signal*. The difficulty of this novel task is that we do not provide any explicit control signal for the object, we only provide a control signal for the human motion, from which the model will need to infer the object dynamics. In addition, the video diffusion model also synthesizes the entire scene, the human motion aligned with the control signal and environmental conditions such as lighting. *To the best of the author's knowledge, this research represents the first attempt to address this specific task*.

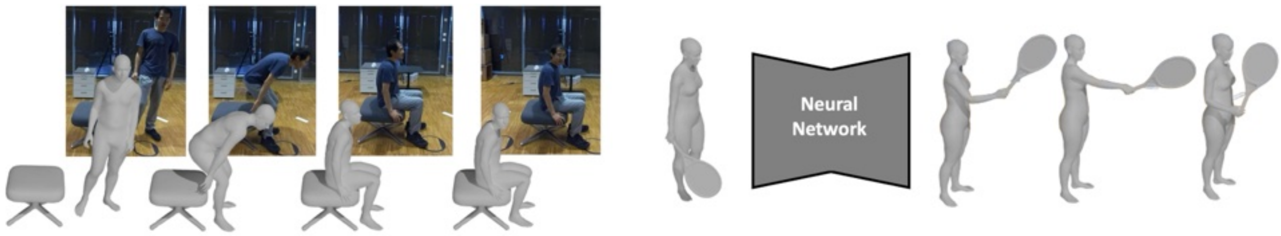
In this master thesis, we utilize the current state-of-the-art image-to-video generation model, Stable Video Diffusion (SVD) [4], as a proxy for a robust world model. This model, trained on an incredibly massive and diverse dataset, has demonstrated the ability to generate intricate videos with realistic dynamics and an understanding of 3D scenes from just a single starting frame. Specifically, we integrate a ControlNet encoder [77] into Stable Video Diffusion (SVD) [4] and fine-tune it using the DexYCB human-object interaction dataset [11]. ControlNet provides a method to preserve the model's weights and introduces an additional trainable copy of its encoder that is integrated with the frozen pipeline through zero convolutions. Through a series of experiments, we investigate the impact of various model architectures and training strategies on video generation performance. Key areas of exploration include comparing two different SVD versions that are fine-tuned on different context lengths to assess the impact on temporal dynamics and object consistency, evaluating both a sparse and dense pixel-aligned control signal, examining the impact of conditioning the ControlNet on CLIP embeddings of the first frame or textual descriptions of objects to enhance object understanding, and manipulating the information flow to the ControlNet to encourage a more generalized control strategy. We employ a range of evaluation metrics, including Fréchet Video Distance (FVD), to assess the quality and realism of the generated videos. Through these experiments, we aim to gain insights into the most effective approaches for image-based HOI synthesis and its degree of world understanding. Please refer to Figure 2 for a visual demonstration of our method.

1.4 Outline

This master thesis is outlined as follows. In section 2, we start off with analysing the literature on 3D HOI synthesis and video generation models. In section 3, we provide extensive easy-to-understand preliminaries on the concept of diffusion models, the extension from image diffusion models to video diffusion models and ControlNet. The concepts outlined there are crucial to understand our method. In Section 4, we explain how we combine SVD and ControlNet, present our control signal and introduce our data processing pipeline. Our experimental setup, evaluation metrics and analysis of results are presented in Section 5. Lastly, in Section 6, we present our conclusion, considerations for future work and an outlook on the future of HOI synthesis.

2 Related Work

2.1 Human-Object Interaction



(a) Capture 3D data **(b)** Generate 3D human-object interaction

Figure 3: Human-object interaction. Examples of (a) 3D human-object interaction capture from multi-view videos [32] and (b) training a model to generate a realistic human-object interaction given a 3D human and a 3D object.

Existing 3D captured HOI datasets are relatively small and lack diversity in the natural human-object interactions they capture, making training large deep neural networks on them difficult. Firstly, interactions involving objects that are impractical to create a 3D version of, or those that are difficult to capture in a lab environment, are often not found in any dataset. Moreover, while some datasets include objects with specific real-world purposes, such as a drill, many fail to capture interactions where these objects are used according to their intended function [11, 25], in this case drilling, or if they do, these interactions are not depicted in natural scenes [3, 22, 32]. Most 3D HOI datasets also do not account for non-rigid deformations that occur when interacting with soft or pliable objects, as these are challenging to capture with a 3D point tracking system [75]. Furthermore, many 3D HOI datasets are specialized towards specific tasks within HOI generation, such as grasping objects but not navigating towards them, or generating whole body motions but no fine-grained finger motions, etc. This specialization, combined with the inherent complexity of human-object interactions, often results in methods that use different models for each stage of the interaction. Lastly, if the goal is to achieve a photo realistic end-product of a human-object interaction, these methods still rely on an explicit 3D scene and depend on rendering engines to model lighting, reflectivity, and texture. Since many 3D human models do not simulate the movement of hair or clothes during interactions, these particularly difficult elements also need to be modeled separately, which can diminish the overall realism of the interaction.

One intuitive approach to address this lack of 3D ground truth data is to train models to estimate 3D human and object poses directly from in-the-wild images or videos. However, this presents significant challenges. Many methods for 3D human pose estimation first detect 2D keypoints in images and then "lift" these into 3D space [12, 33, 41, 49, 59]. While this approach can provide reasonable estimates of skeletal structure, keypoints alone often lack detailed information about contact regions between human and object that are necessary for understanding human-object interactions. To overcome these limitations, some methods focus on estimating the parameters of a 3D body model and mesh from the detected 2D keypoints [6, 34]. This approach not only provides a prior for human shape, potentially reducing the solution space of possible poses, but also results in a more complete 3D representation of the human body. Accurate 3D object pose estimation remains difficult, with techniques like template matching, feature-based methods [60, 69, 74], and deep learning approaches [31, 48] each facing their own limitations. Importantly, for humans we can capture a statistical shape prior, which is something that does not work for objects in general. Additionally, ensuring accurate contact registration between the estimated human and object meshes is a significant hurdle, especially in occluded regions. Pioneering research by Hasson et al. [26] highlighted the potential of joint reconstruction models that simultaneously estimate both hand and object poses. Such an approach leverages physical interaction constraints such as non-penetration and contact consistency to reduce the solution space for both the human hand and the object, potentially improving accuracy. In general, while all of these methods are promising for static images, extending them to video data introduces additional complexities such as temporal dependencies, motion blur, and temporary occlusions.

2.2 Video Generation

Video generation aims to create realistic or stylized video content automatically. This can include generating completely new videos, enhancing existing ones, or creating visual effects that would be difficult or time-consuming to produce manually. Whether the input modality is text, image or noise, the model's main mechanism lies in its ability to find spatial and temporal relationships within videos [15]. Much like the saying: "Rome wasn't built in a day", the evolution of video generation field consists of a gradual accumulation of sophisticated innovations. Early approaches focused on establishing the foundational concepts of video generation, often relying on smaller datasets and manual feature engineering. However, the advent of powerful models like SORA, a transformer-based diffusion model, has significantly elevated the field's capabilities [1].

SORA's success, while groundbreaking, should not overshadow the diverse approaches that have shaped video generation. Recurrent Neural Networks (RNNs), generative auto-encoders, GANs, auto-regressive transformers, diffusion models, latent diffusion models, attention mechanisms, CLIP, 3D convolutions and skip connections, are just a few of the key components that have paved the way for today's advanced models. Not long ago, these techniques were considered the state-of-the-art. The integration of these techniques, coupled with the availability of large-scale, high-quality image and video datasets, has enabled a shift towards unsupervised learning methods. These methods leverage the power of computation and self-organization to extract patterns and relationships from data, reducing the reliance on human-engineered features.

The combination of these advancements has propelled video generation beyond simple image manipulation, enabling the creation of dynamic and high-quality video content. In the following section, we will delve deeper into the foundational concepts that underpin modern video generation, exploring how they have shaped the field's trajectory and set the stage for future innovations.

Variational Autoencoders (VAEs). VAEs are deep learning models that generate new content by regularizing the latent space of an autoencoder [37]. Regularization ensures that the latent space is organized in a way that allows for meaningful content generation. This is done by encoding inputs as distributions over the VAE's latent space, and by enforcing these distributions to be close to a standard normal distribution. By sampling points from this regularized latent space and decoding them, VAEs can generate new and diverse content. Recently, Vector Quantized Variational Autoencoders (VQ-VAE) extended the capabilities of traditional VAEs by introducing a discrete latent representation instead of a continuous normal distribution [64]. It uses vector quantization, where the encoder outputs are mapped to the nearest vector in a predefined codebook. The decoder then reconstructs the input from these quantized vectors.

Early work, such as Sync-DRAW [44], demonstrated the potential of combining VAEs with recurrent attention mechanisms for video generation. This approach effectively learned a global latent representation of a video, with frame-level focus using the recurrent attention mechanism.

However, sampling a single latent representation from the prior distribution for the entire video meant that all variability and movement within the video had to be encoded in a single latent vector. This severely restricted the model's ability to capture the full range of potential future outcomes. Subsequent research, such as SV2P, addressed this by sampling from the same fixed prior once per frame [2], which allowed the model to generate a variety of different continuations for a given video. Later this was extended in SVG [17] towards learning a prior that is conditioned on the past frames of the video, enabling the model to capture the specific patterns of uncertainty. Still, VAE approaches suffered from producing blurry predictions, which was later identified as a sign of underfitting and was alleviated by increasing the expressiveness of the latent distributions and using a higher capacity likelihood model [10]. Finally, VideoGPT [76] combined the power of VAEs with the Transformer architecture [65]. The work proposes to use a VQ-VAE to learn a compressed representation of videos and then use a transformer to model the temporal dependencies between frames.

Generative Adversarial Networks (GANs). GANs consist of two neural networks, a generator, and a discriminator, engaged in a mini-max game [24]. The generator receives a random noise vector from a latent space as input and learns to map it onto a data instance in the target domain, such as an image. It aims to produce outputs that are statistically indistinguishable from the real data distribution, while the discriminator's objective is to differentiate between authentic training examples and synthetic instances produced by the generator. The discriminator receives either a real data sample or a generated sample and outputs a probability indicating its confidence in the sample's authenticity. Through this iterative adversarial training process, GANs have demonstrated exceptional capacity for modeling complex, high-dimensional data distributions and generating realistic samples across diverse domains [7, 36].

Naturally, the field for video generation has sought to extend the capabilities of GANs to the video domain, which can roughly be divided into four non-exclusive categories. The first category focuses on a multi-stream GAN architecture. For instance, some works use a multi-stream Generator architecture that aims to disentangle the motion and content generation by specializing in different aspects of the video, such as foreground and background [68] or motion and content [55, 57, 71]. Other research noticed that often the Discriminator network did not have the capacity to capture temporal and spatial inconsistencies at the same time and therefore employed a multi-stream Discriminator architecture, such as for motion and content [16, 39, 58, 61, 62, 71] or for motion, content and conditioning alignment [47]. The second category applies a coarse-

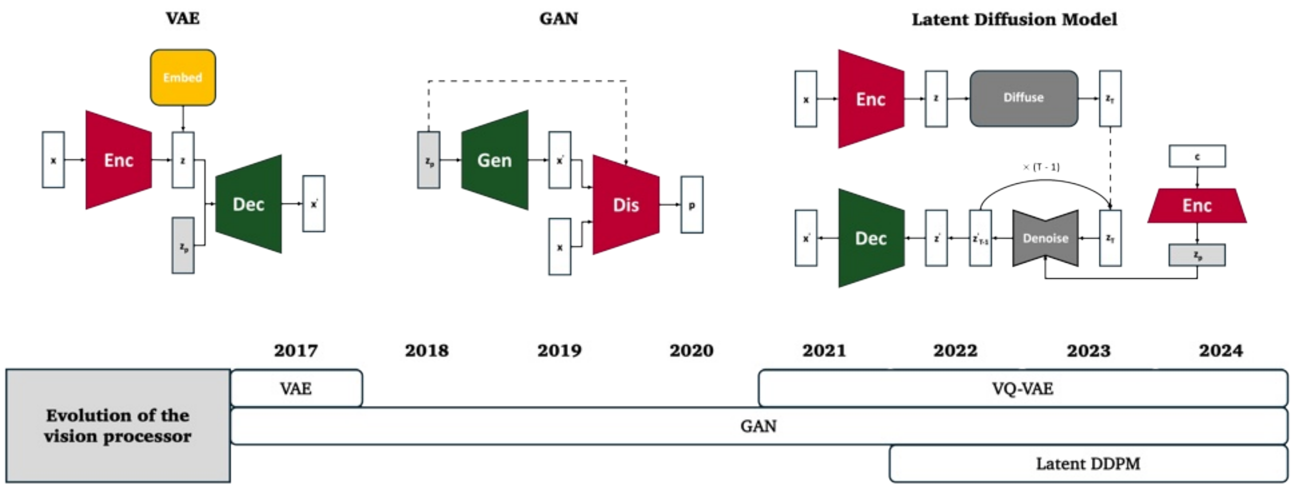


Figure 4: Video Generation Timeline. Evolution of the vision backbone. Figure by [15].

to-fine strategy, starting with low-resolution frames and increasing the resolution to manage complexity and improve the quality of generated videos [8, 19, 38]. The third approach is to find latent space trajectories in the latent space of a pre-trained image generator [61]. This trajectory represents the motion dynamics of the video, and the image generator is used to synthesize the actual video frames based on the latent codes along the trajectory. The final category classifies video generation methods based on their approach to frame generation. Approaches include models that generate a set of frames in one step simultaneously often using a combination of 1D and 2D convolutions, or 3D convolutions [8, 38, 47, 68, 71], models that generate frames concurrently with a shared Generator [16, 39, 61, 62], or models that generate frames autoregressively [58, 66]. Despite these advancements, video generation with GANs still faces challenges such as locality bias and spatial invariance of the CNN backbone [21], mode collapse, particularly in conditional generation settings, and difficulties in scaling to high-resolution and long-duration videos [15, 45].

Diffusion Models (DMs). DMs represent a class of generative models that gradually transform noise into structured data through a series of learnable denoising steps. Initially introduced by Sohl-Dickstein et al. [56], these models operate by first gradually adding noise to a data sample until it is completely randomized, and then learning to reverse this process. Each step of the reverse process involves predicting the original data from the noised version, effectively training the model to denoise the data progressively. This allows DMs to produce high-quality, coherent outputs by refining the output step by step, contrary to the direct generation methods employed by GANs and VAEs. Recent advancements in DMs have significantly improved their efficiency and output quality, enabling them to compete with or even surpass other leading generative models in tasks such as image synthesis, text-to-image generation, and audio synthesis [28, 46].

Bounded Video Generation

One recent study demonstrates how Stable Video Diffusion (SVD) can generate realistic dynamics in a bounded setting without the need for fine-tuning [23]. It merges forward and backward generation paths, using the initial and final frames as conditions, where the model needs to hallucinate the motion and dynamics in between these two frames. This capability demonstrates that the model does more than generating a bag of images — it is able to predict and synthesize plausible future states, integrating complex motions and 3D-consistent views guided by the context provided by the bounding frames. While bounded generation showcases the capabilities of SVD, using this method for HOI synthesis has clear practical considerations. First, although the first and the last frame act as an effective constraint on the motion in the scene, we do not have precise control over the motion path. Secondly, due to the stochasticity in the generation of forward and backward passes, the method has significant variability in the motion paths for given start and end frames, which can also result in videos that are not realistically fused. Especially, human-object interactions with considerable displacements seem difficult for this method to fuse correctly, as we discovered through empirical analysis. See an example in Fig. 5.

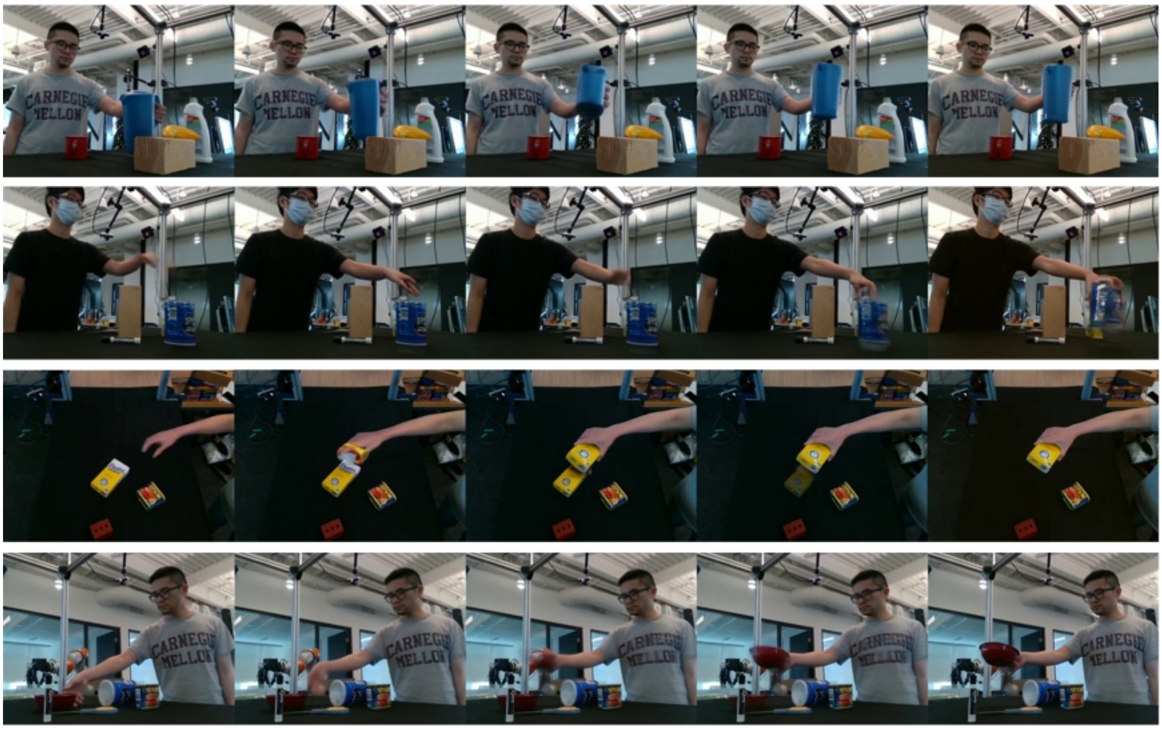


Figure 5: Bounded generation. Success and failure cases for human-object interaction. Row 1: successful case, Row 2: failure case because the hand disappears, Row 3: failure case because the object appears and disappears, Row 4: failure case because the object appears and disappears.

3 Preliminaries

3.1 Diffusion Models

Diffusion Models (DMs) employ two processes. The *forward process* gradually adds random noise to a data sample, effectively pushing the sample away from the distribution of the original data $p_{\text{data}}(\mathbf{x})$. The *reverse process* performs the opposite action, reducing the noise level and nudging the sample back toward the data distribution. At a very high noise level, denoted as $\sigma_{\max}$, the distribution approximates an isotropic normal distribution with mean zero and variance $\sigma^2_{\max}$. At inference time, this allows us to start with a noisy sample drawn from this distribution, and then iteratively denoise it to gradually uncover a noise-free sample that is indistinguishable from $p_{\text{data}}(\mathbf{x})$ [28], see Figure 6. For an intuitive visual explanation of how diffusion models work, I advise to visit Erdem's blogpost [20].

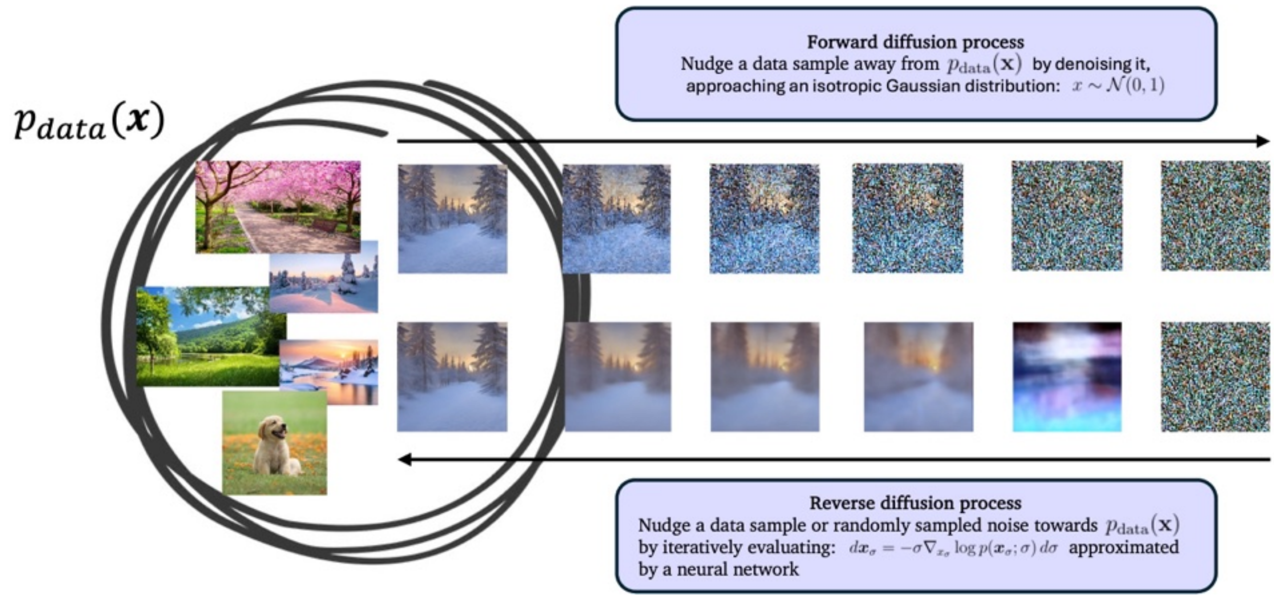


Figure 6: Diffusion process. Simple visualisation of the diffusion process.

To continuously increase or reduce the noise level of an image, moving either to or away from $p_{\text{data}}(\mathbf{x})$, we can either use a stochastic differential equation (SDE) or an ordinary differential equation (ODE). SDE-based diffusion models incorporate randomness in the reverse diffusion process, which makes the backward path inherently stochastic. This allows for better exploration of the image space, leading to more realistic and varied outputs. However, it often comes at a computational cost. In ODE-based diffusion models on the other hand, we do not incorporate randomness in the reverse process. This means that the only source of stochasticity is the initial noise, with the backward path being deterministic. When we refer to diffusion models in this preliminary section, we refer to ODE-based diffusion models.

As mentioned before, in ODE-based diffusion models the reverse process is governed by a probability flow ordinary differential equation:

$$\mathrm{d}\mathbf{x}_\sigma \;=\; -\sigma\nabla_{x_\sigma}\log p(\mathbf{x}_\sigma;\sigma)\;\mathrm{d}\sigma. \tag{1}$$

As σ approaches zero, the process reveals $\mathbf{x}_0$, which ideally corresponds to a sample from the original data distribution $p_{\text{data}}(\mathbf{x})$. In practice, solving this equation directly is challenging, so in the reverse diffusion process we do it step-by-step, numerically. This requires evaluating $\nabla_x \log p(\mathbf{x};\sigma)$ at each step, known as the score function, which tells us how much and in which direction to adjust our noisy sample to make it less noisy.

What gave rise to the field of diffusion models, is the idea that we could approximate $\nabla_x \log p(\mathbf{x};\sigma)$ using a neural network D_θ parameterized by weights θ in a denoising task:

$$\theta = \arg\min_\theta \mathbb{E}_{\mathbf{s}\sim p_{\text{data}},\sigma\sim p_{\text{train}},\mathbf{n}\sim\mathcal{N}(\mathbf{0},\sigma^2\mathbf{I})}\|D_\theta(\mathbf{s}+\mathbf{n};\sigma)-\mathbf{s}\|_2^2, \tag{2}$$

where s is an image sampled from the ground truth data distribution p_{data}, $\mathbf{n}$ is noise and $p_{\text{train}}(\sigma)$ is a distribution from which we sample σ. Once D_θ is fully trained, we can then use it to estimate the score function up to approximation errors:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x};\sigma) \approx (D_\theta(\mathbf{x};\sigma)-\mathbf{s})/\sigma^2. \tag{3}$$

Training challenges. Training the neural network D_θ directly according to Equation 2 often results in subpar performance, as empirically reported by [28]. This is because the input $\mathbf{x} = \mathbf{s}+\mathbf{n}$ is a combination of a clean signal s and a noise component $\mathbf{n} \sim \mathcal{N}(\mathbf{0},\sigma^2\mathbf{I})$ of which the magnitude varies immensely depending on the sampled noise level σ. In order to stabilize the training process of diffusion models, the field applied various well-known best practices, such as ensuring consistent magnitudes for inputs and outputs, as well as managing gradient variances. One straightforward solution is to normalize the input with σ, train a model F_θ to predict the full noise $\mathbf{n}$ scaled to unit variance, and reconstruct the signal via $D_\theta(\mathbf{x};\sigma) = \mathbf{x} - \sigma F_\theta(\cdot)$. However, the disadvantage of this approach is that any mistake made by F_θ will be heavily amplified by σ, which is especially detrimental at the start of the denoising process. In such cases, the model is likely to spend multiple iterations trying to fix its previous mistake. The literature slowly moved to methods that adaptively mix signal and noise. This allows the model at high noise levels to denoise aggressively, while at low noise levels it can retain the largely denoised signal.

EDM framework. In an effort to standardize these techniques, Karras et al. [35] proposed a unified framework that introduces a set of preconditioning functions:

$$D_\theta(\mathbf{x};\sigma) = c_{\text{skip}}(\sigma)\mathbf{x} + c_{\text{out}}(\sigma)F_\theta(c_{\text{in}}(\sigma)\mathbf{x};c_{\text{noise}}(\sigma)). \tag{4}$$

where $\mathbf{x}$ is a noisy data sample $\mathbf{x} = \mathbf{s}+\mathbf{n}$, $\mathbf{s}$ is a clean data sample, $\mathbf{n} \sim \mathcal{N}(\mathbf{0},\sigma^2\mathbf{I})$ is the added noise, $\sigma \in \mathcal{N}(P_{\text{mean}}, P_{\text{std}}^2)$ is the sampled noise level and F_θ is parameterized by a neural network.

The primary role of $c_{\text{in}}(\sigma)$ is to scale the input data $\mathbf{x}$ such that the network receives inputs within a consistent and manageable range of values across different noise levels σ. This scaling is essential because (1) at high noise levels, the variance in $\mathbf{x}$ leads to a broad range of input values, which can cause unstable gradients during training, and (2) at low noise levels, the variance in $\mathbf{x}$ is much smaller, such that the network is likely to be insensitive to variations in the input. The role of $c_{\text{noise}}(\sigma)$ is to inform the neural network about the current noise level. We provide $c_{\text{noise}}(\sigma)$ as conditioning, which allows the network to adapt its denoising strategy, effectively applying aggressive denoising when noise is high and conserving details when noise is low. After the input has been passed through the network, the role of $c_{\text{out}}(\sigma)$ is to scale the output to ensure it matches the expected scale of the target data. Lastly, $c_{\text{skip}}(\sigma)$ controls the extent to which the original input is retained in the final output. Often $c_{\text{skip}}(\sigma)$ is implemented as an inversely proportional function. This means that at high levels of noise σ, $c_{\text{skip}}(\sigma)$ allows the reverse diffusion process to rely more on the network's noise filtering, while at low levels of noise σ, $c_{\text{skip}}(\sigma)$ preserves more of the original input $\mathbf{x}$.

Finally, if we take a weighted expectation of Equation 2 over the noise levels, we get the overall training loss:

$$\mathbb{E}_{\sigma,\mathbf{s},\mathbf{n}}\left[\lambda(\sigma)\;\|D(\mathbf{s}+\mathbf{n};\sigma)-\mathbf{s}\|_2^2\right], \tag{5}$$

where $\sigma \sim p_{\text{train}}$, $\mathbf{s} \sim p_{\text{data}}$, and $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$. The probability of sampling a given noise level σ is given by $p_{\text{train}}(\sigma)$ and the corresponding weight is given by $\lambda(\sigma)$. Combining Equation 5 and Equation 4, the loss can be expressed with respect to the raw network output F_θ:

$$\mathbb{E}_{\sigma,\mathbf{s},\mathbf{n}}\Big[\underbrace{\lambda(\sigma)\;c_{\text{out}}(\sigma)^2}_{\text{effective weight}}\Big\|\underbrace{F_\theta\big(c_{\text{in}}(\sigma)\cdot(\mathbf{s}+\mathbf{n});c_{\text{noise}}(\sigma)\big)}_{\text{network output}}-\underbrace{\tfrac{1}{c_{\text{out}}(\sigma)}\big(\mathbf{s}-c_{\text{skip}}(\sigma)\cdot(\mathbf{s}+\mathbf{n})\big)}_{\text{effective training target}}\Big\|_2^2\Big]. \tag{6}$$

In order to balance the loss function over the entire noise range, the authors set $\lambda(\sigma) = c_{\text{out}}(\sigma)^2$, which cancels out the effective weight.

Noise sampling. After training, Karras et al. [35] observed that the per-sample loss was the highest for both high and low noise levels. The authors hypothesize that at high levels of noise the neural network has a difficult time to discern the exact noise within an incredible noisy sample, while at low levels of noise the model must perform very precise adjustments to remove the incredibly small levels of noise that are hard to distinguish from the signal. Essentially, both of these edge cases are incredibly difficult: one ill-posed and the other extremely sensitive. To avoid wasting considerably resources on these inherently difficult samples during training, the authors therefore advise to sample the noise level σ from a simple log-normal distribution, centered around medium noise levels.

Architecture. While many leading-edge diffusion models employ $F_\theta(\cdot)$ based on a transformer architecture [65], almost all of the publicly available models employ some type of modified U-Net [54], since they are less compute-heavy to train. A U-Net generally consists of an encoder that progressively downsamples the input image to capture high-level features, a bottleneck, and a decoder that upsamples the input back to the original size, refining the image details. To capture fine-grained details and spatial information, the U-Net uses so-called skip connections, in which the feature maps from the downsampling layes are directly concatenated to the feature layers of the upsampling layers. Lastly, many U-Net variants use attention modules in the encoder and decoder to enable the network to focus on relevant image regions at different time steps during the diffusion process.

Latent Diffusion. One key innovation in diffusion models was to apply the forward and reverse diffusion process not directly to the raw pixel values of an image, but to a compressed latent space of a Variational Auto-encoder [52]. Here, the encoder and decoder of a pre-trained, frozen VAE are used to first encode an input image to the latent space, and at the end of the diffusion process back from latent space to image space. Please refer to Figure 7 for an architectural overview. Applying the diffusion process in this manner significantly reduces the computational burden, making it more feasible for training on, and generating, high-quality images. Moreover, by leveraging the VAEs latent space, which captures the most salient image features, latent diffusion often results in not only efficient generation but also an improvement in overall output quality.

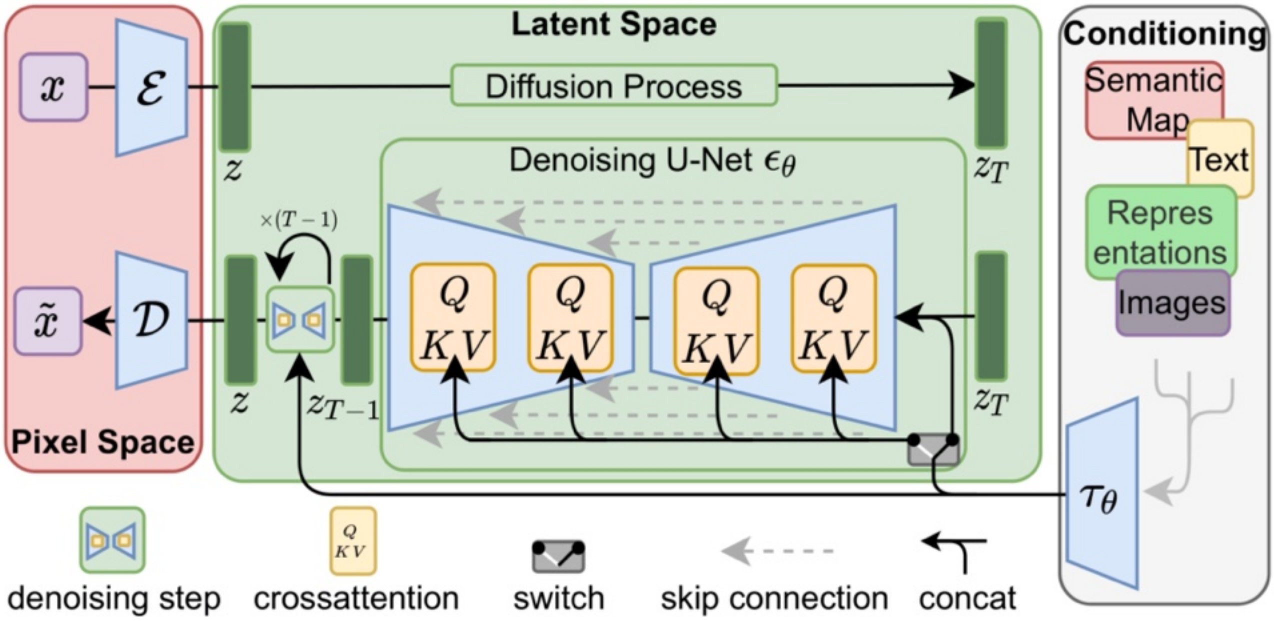


Figure 7: Latent Diffusion Model. In the latent diffusion process, we first encode an image from pixel space (red) to latent space (green) and then apply the diffusion process. Afterwards, we can retrieve a final output image by decoding the denoised latent back to image space. Figure by Rombach et al. [52].

Guidance. Guidance is a technique used to steer the backward diffusion process towards a specific conditioning signal $\mathbf{c}$ at *inference time*. In classifier guidance [18], we use the gradients of a trained classifier to steer the generation towards a specific class. For instance, using the gradients of a cat classifier we can guide the denoising process towards the distribution of cat samples $p_{\text{cat}}(\mathbf{x})$. Two apparent issues with classifier guidance is that (1) the number of classes we can steer the generation towards is severely limited by the classes the classifier is trained on, and (2) for accurate guidance the classifier needs to be trained on noisy data. Classifier-free guidance [29] improves upon these challenges by taking a different approach. The main idea is to blend the outputs from a conditional model and an unconditional model as follows:

$$D_w(\mathbf{x}; \sigma, \mathbf{c}) = wD(\mathbf{x}; \sigma, \mathbf{c}) + (1 - w)D(\mathbf{x}; \sigma), \tag{7}$$

where $w \geq 0$ represents the *guidance strength*. The unconditional term $D(\mathbf{x}; \sigma)$ is often trained simultaneously with the conditional model by intermittently replacing the conditioning signal $\mathbf{c}$, with a null vector [29]. This way the model learns a robust representation that can generalize well across conditioned and unconditioned scenarios. This approach not only enhances the model's flexibility, but also its ability to generate high-fidelity outputs under various conditions using the guidance strength.

3.2 Stable Video Diffusion

Diffusion models have revolutionized image generation, achieving state-of-the-art (SOTA) results. The logical next step is to extend this success to videos. However, videos pose unique challenges due to their added temporal dimension, making large-scale dataset collection and architectural design more complex.

Architecture. The core idea of Blattmann et al. [4] for efficiently training a video model involves re-purposing a pre-trained text-to-image generator and extend it to the realm of video.

Here, the layers of the pre-trained image diffusion model are denoted as *spatial* layers l_θ^i with layer index i, which are interleaved with randomly initialized *temporal* neural network layers l_ϕ^i. Whereas the spatial layers process each frame independently, the L additional temporal layers $\{l_\phi^i\}_{i=1}^L$ process the frames over the temporal dimension. The authors introduce two types of temporal layers, that together form one temporal block: (1) residual layers based on 3D convolutions, (2) and temporal attention layers. Sinusoidal embeddings [65] provide the model with a positional encoding for time, allowing it to differentiate the order of the frames. The full model $\mathbf{f}_{\theta,\phi}$ is thus the combination of the spatial layers (595M parameters) and temporal layers (656M parameters) with a total size of 1521M parameters. The advantage of training a video diffusion model in this way is that huge image datasets can be used to pre-train the spatial layers, while the temporal layers can be trained with comparatively smaller sized high-quality video datasets. Please refer to Figure Figure 8 for a visual overview of the proposed denoising U-Net that stands at the basis of Stable Video Diffusion.

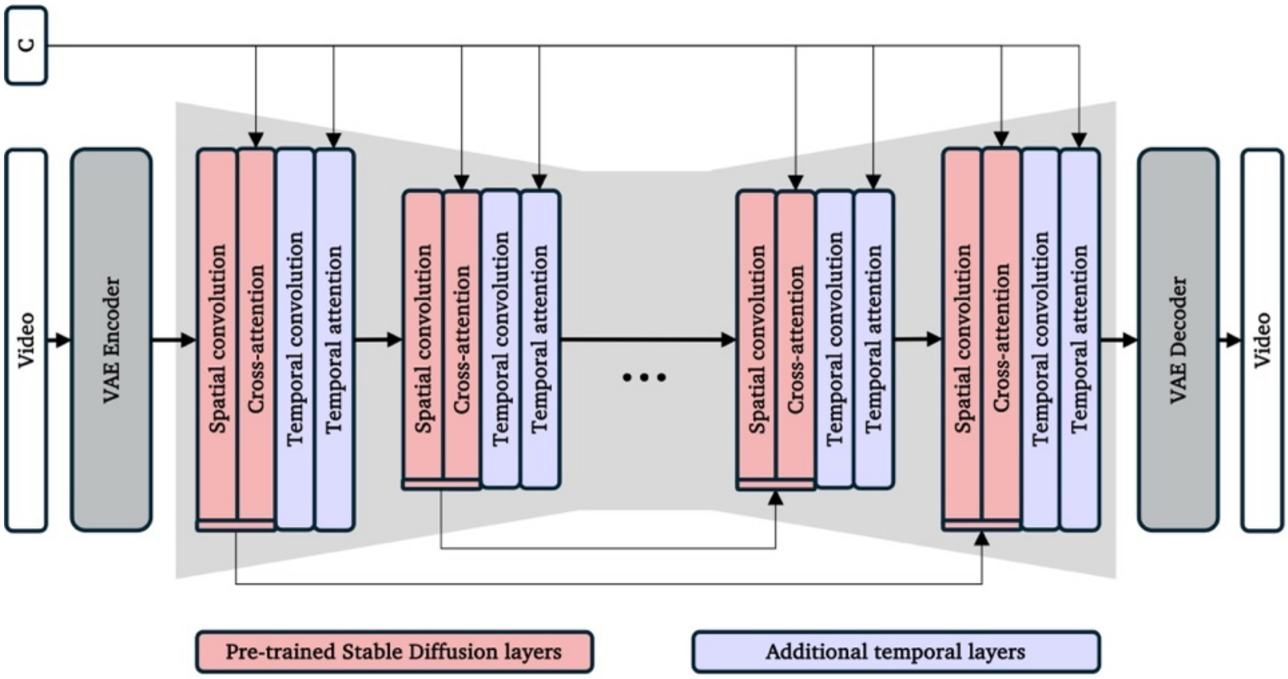


Figure 8: Video U-Net. The Denoising U-Net (i.e., SVD), consists of pre-trained spatial layers (shown in red) and additional temporal layers (shown in blue).

The U-Net of SVD receives a batch of frame-wise encoded input videos $\mathcal{E}(\mathbf{x}) = \mathbf{z} \in \mathbb{R}^{B\times T\times C\times H\times W}$, where B is the batch size, T is the number of frames in the video, C is the number of latent channels and H and W are the spatial latent dimensions. The spatial layers process the video as a batch of independent images by treating the temporal dimensions as an extension of the batch dimension. Next, for each *temporal mixing layer* l_ϕ^i, SVD reshapes back to the video dimensions as follows (using `einops` notation):

$$\text{Spatial layers:} \quad \mathbf{z}' \leftarrow l_\theta^i(\mathbf{z}, \mathbf{c}) \tag{8}$$

$$\text{Reshape:} \quad \mathbf{z}' \leftarrow \texttt{rearrange}(\mathbf{z}',\ \texttt{(b t) c h w} \rightarrow \texttt{b c t h w}) \tag{9}$$

$$\text{Temporal layers:} \quad \mathbf{z}' \leftarrow l_\phi^i(\mathbf{z}', \mathbf{c}) \tag{10}$$

$$\text{Reshape:} \quad \mathbf{z}' \leftarrow \texttt{rearrange}(\mathbf{z}',\ \texttt{b c t h w} \rightarrow \texttt{(b t) c h w}), \tag{11}$$

Training. Stable Diffusion 2.1 is first transformed into a text-to-video model throughout multiple training stages, as detailed in Table 1. The model receives a noised video sample, as well as a conditional text prompt that is first converted into a CLIP embedding representation before being supplied to the cross-attention and temporal attention layers. In addition, it receives the FPS of the video, as well as the motion ID, which is an integer representing the amount of motion present in the video based on optical flow. A higher motion ID results in more dynamic pixel behavior, and vice versa. In this way the model at inference time can generate videos from lower FPS to higher FPS, and videos with small motions or large motions, all based on a textual prompt.

Stage	Type	Resolution	Iterations	Purpose
1	Image	512×512	1k	Adapt to new preconditioning functions [35] and a continuous noise schedule
2	Image	256×384	30k	Adjust to new resolution
3	Video	256×384	150k	Insert temporal layers and train on videos
4	Video	320×576	100k	Increase resolution
5	Video	320×576	50k	Turn the text-to-video model into a image-to-video model
6	Video	576×1024	50k	Increase resolution

Table 1: SVD is trained in several stages that first introduce a new noise schedule and then slowly increase the spatial resolution and video quality over time.

The eventual image-to-video diffusion model is acquired during training stages 5 and 6, in which the text-to-video model is fine-tuned to accept a conditioning image as input rather than a textual prompt. This is possible since the text was first encoded to a so-called CLIP embedding [51], where CLIP is a joint image and text embedding model that maps text and images to the same embedding space. This means that for instance an image of a banana and the sentence "an image of a banana" will be encoded with very similar embeddings. Therefore, the authors simply replace the text CLIP embedding with the CLIP embedding of the conditioning frame. In addition, the authors concatenate the latents of the conditioning frame to the input of the U-Net, by simply repeating the conditioning frame across the time axis: $\mathcal{E}(\mathbf{x}) = \mathbf{z} \in \mathbb{R}^{B\times 2T\times C\times H\times W}$. Before transforming the conditioning frame from image space to latent space, the authors add a tiny bit of noise to the conditioning frame to make the model robust for diverse image qualities at inference time. Here, the CLIP embedding of the conditioning frame provides high-level semantic information of the scene, whereas the noise-augmented concatenated conditioning frame provides low-level, precise information. This allows the model to generate videos that are not only semantically similar to the conditioning frame, but actually are a continuation of the conditioning frame. The authors provide two image-to-video models: one trained with a context length of 14 frames, and one another trained with a context length of 25 frames. Both versions are trained starting from the same base model with identical hyperparameters, except for the context length. The authors supply both models on their HuggingFace page, including their inference code.

At inference time, given an initial input frame $\mathbf{f} \in \mathbb{R}^{1\times C\times H\times W}$ and a pure noise sample $\mathbf{n} \in \mathbb{R}^{T\times C\times H\times W}$, SVD generates a video sequence of T frames. This sequence is constructed through the reverse denoising diffusion process where, at each denoising step, the conditional U-Net is used to iteratively denoise $\mathbf{n}$.

Motion ID. One critical aspect is the influence that the motion ID has on the generated video. Empirically, we notice that a lower motion ID correlates to video generations in which there is movement *within* the scene, whereas a higher motion ID correlates to movement *of* the scene, representing camera motion, see Figure 9. We hypothesize that since the motion ID is based on the optical-flow measure, the model learns a spurious correlation between a large motion ID and camera movement, which tends to have a large pixel movement and hence optical-flow score. Choosing an appropriate motion ID is therefore not straightforward, since it refers to two vastly different concepts of motion. Interestingly, in a static scene, a large motion ID may lead to excessive camera motion or unnatural moving objects in the background, while a motion ID of zero will simply generate a repetition of the conditioning frame. *This means that the amount of motion in the scene is not easily controllable out-of-the-box.* We explore the entire range of the Motion ID for various starting frames in which a human holds an object and empirically find that a motion ID of 40 seems to correlate the most to a generation with a human interacting with the object and minimal camera movement. Please refer to Figure 9 for an illustration of the difference of generation with a low and high motion ID.

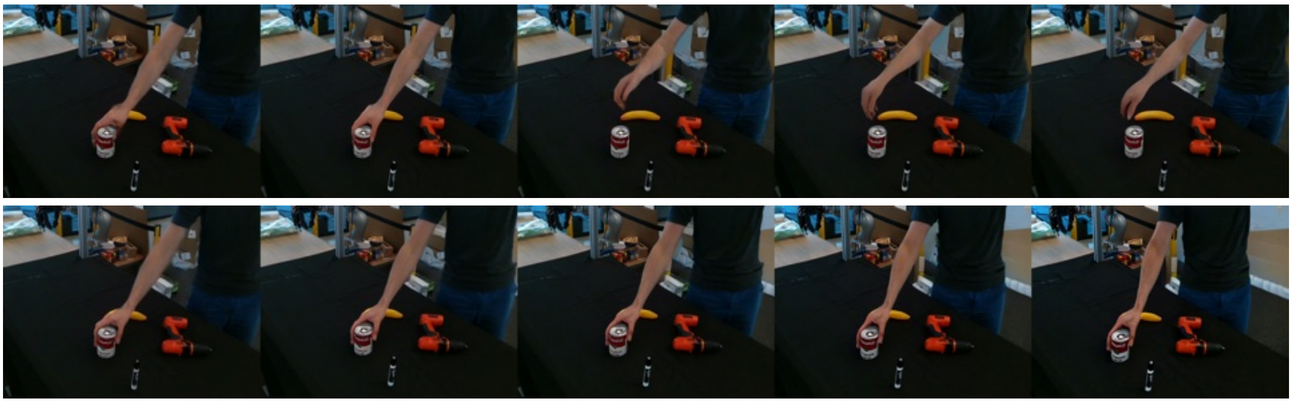


Figure 9: Exploration of motion ID. Above: motion ID 40, below: motion ID 140. Notice that a low motion ID correlates to movement in the scene and high a motion ID to camera movements.

Furthermore, we notice that SVD generally does not generate videos with human-object interactions if the conditioning frame does not hint at this severely. For instance, in the teaser examples shown in Section 1 there is clear human-object interaction within the conditioning image, which increases the probability that SVD will generate a video of the human interacting with the object. Despite this, there is still a tendency of the model to generate videos with camera movement. We also notice that the model is rather sensitive to the sampled starting noise at inference time, where for an identical conditioning frame but for noise sampled with a different random seed, the model generates videos with completely different motions. This means that at inference time, researchers would do good in taking the mean on a per-sample basis for a number of different random seeds.

3.3 ControlNet

Out-of-the-box, text-to-image diffusion models provide limited control over the precise spatial composition of the generated images, since only a coarse textual description is provided. However, several downstream applications require the ability to precisely describe the spatial composition of the output, using some type of control signal such as edge maps, depth maps, or even human skeletons. One way to embed such spatial control is by finetuning a diffusion model to accept both a textual prompt and a control signal. In this way, the text determines the semantics of the scene, while the control signal can manipulate the composition of these semantics. However, datasets that have such paired image and control data are often limited in size, introducing the risk of overfitting, diminished generalizability, or even catastrophic forgetting. ControlNet [77] tackles this challenge with an end-to-end neural network architecture designed to learn conditional controls for large pre-trained diffusion models without losing their generative capabilities.

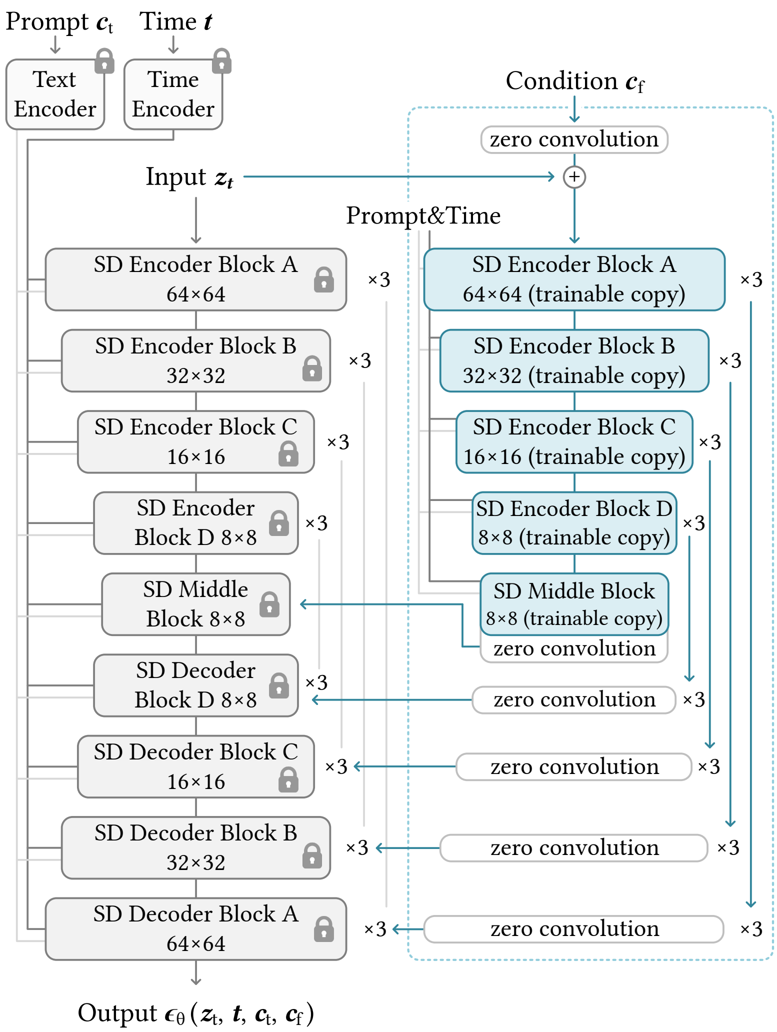


Figure 10: ControlNet Architecture. Stable Diffusion's U-Net architecture connected with a ControlNet on the encoder blocks and middle block. The locked, gray blocks show the structure of Stable Diffusion 2.1. The trainable blue blocks and the white zero convolution layers are added to build a ControlNet. Figure by [77].

The key idea of ControlNet is to freeze the parameters of a pre-trained model, and extend the architecture with an *additional trainable encoder*, initialized using the same weights as the frozen encoder. An illustration of the architecture is shown in Figure 10. Whereas the frozen encoder-decoder architecture only receives the noisy sample, the additional encoder receives both the noisy sample and the control signal. The goal of the trainable encoder is to learn the relationship between the control signal and the noise input, and supply the layers of the original frozen network with the necessary information to process the input accordingly. To facilitate this one-directional information exchange (from the additional trainable encoder to the frozen pipeline) the two are connected via skip connections at each encoder block as well as their middle layers, please refer to Figure 10. In practice the skip connection features of the frozen encoder and the ControlNet encoder are simply concatenated, but other possibilities have been explored in the literature [50]. Key to ControlNet is the idea of using *zero convolutions*: 1×1 convolutional layers $\mathcal{Z}(\cdot;\cdot)$ where both weights and biases are initialized to zero. The output of the zero convolution block is simply added to the corresponding block in the frozen model, thus allowing the ControlNet encoder to modulate the information flow, without hurting the performance of the original model at the beginning of training. As training progresses, these zero convolution layers gradually learn to incorporate the conditional information.

Specifically, suppose $\mathcal{F}(\cdot;\Theta)$ is a trained neural block with parameters Θ, that transforms an input feature map $\boldsymbol{x}$ into another feature map $\boldsymbol{y}$:

$$\boldsymbol{y} = \mathcal{F}(\boldsymbol{x};\Theta). \tag{12}$$

In the setting of ControlNet, $\boldsymbol{x}$ and $\boldsymbol{y}$ are usually 2D feature maps, i.e. $\boldsymbol{x} \in \mathbb{R}^{h \times w \times c}$ with $\{h, w, c\}$ the height, width, and number of channels in the map, respectively. The corresponding ControlNet block then computes:

$$\boldsymbol{y}_{\mathrm{c}} = \mathcal{F}(\boldsymbol{x};\Theta) + \mathcal{Z}(\mathcal{F}(\boldsymbol{x} + \mathcal{Z}(\boldsymbol{c};\Theta_{\mathrm{z1}});\Theta_{\mathrm{c}});\Theta_{\mathrm{z2}}), \tag{13}$$

where $\boldsymbol{y}_{\mathrm{c}}$ is the output of the ControlNet block, and $\mathcal{Z}$ is the zero convolution layer. The second part of the equation shows that both the control signal $\boldsymbol{c}$, as well as the entire output of the block $\mathcal{F}(\boldsymbol{x} + \mathcal{Z}(\boldsymbol{c};\Theta_{\mathrm{z1}});\Theta_{\mathrm{c}})$ are passed through a zero convolution. Importantly, since both the weight and bias parameters of a zero convolution layer are initialized to zero, this effectively nullifies the influence of the control signal at the start of training:

$$\boldsymbol{y}_{\mathrm{c}} = \boldsymbol{y}. \tag{14}$$

To add ControlNet to latent diffusion models, the control signal is first encoded into the same feature space size of the denoising U-Net. The authors of ControlNet propose to use a tiny network $\mathcal{E}(\cdot)$ to encode an image-space condition $\boldsymbol{c}_{\mathrm{i}}$ into a feature space conditioning vector $\boldsymbol{c}_{\mathrm{f}}$ as

$$\boldsymbol{c}_{\mathrm{f}} = \mathcal{E}(\boldsymbol{c}_{\mathrm{i}}). \tag{15}$$

where the conditioning vector $\boldsymbol{c}_{\mathrm{f}}$ is regarded as a latent representation of the control signal.

The weights of ControlNet are initialized using the weights of the U-Net rather than random weights, since Zhang et al. found that starting the ControlNet off from the generative capabilities of the original model, produces better results. To close off, ControlNet has become the defacto method for control over diffusion models, since it provides a method that preserve the production-ready model trained with billions of images, while the trainable copy reuses such large-scale pre-trained model to establish a deep, robust, and strong backbone for handling diverse input conditions.

4 Method

Our goal is to generate videos of HOI given a control signal for human motion and an initial frame. In order to achieve this goal, we explore the use of a video generation model, since they have the potential to be a world model. In particular we opt to use SVD [4], which is the current state-of-the-art in generative video generation. In order to take control of its implicit latent space and avoid catastrophic forgetting, we freeze its weights and employ an additional ControlNet encoder that maps a control signal to the frozen decoder. We train the ControlNet encoder on DexYCB, a human-object interaction dataset. In this section, we detail the methodologies employed to tackle the complex task of human-object interaction synthesis from a single image input guided by a control signal. Specifically, we begin by explaining the novel task that we address in Section 4.1, and the design of the control signal in Section 4.2. In Section 4.3 we detail our architecture design and detailed training steps. Finally, in Section 4.4 we explain the details of the dataset employed, as well as the pre-processing steps.

4.1 Task

In this master thesis, we introduce the task of human-object interaction synthesis using a single image input and a *control signal* that guides a *human* to manipulate an *object* within a scene. Please refer to Figure 2 for a visualization.

Let $\boldsymbol{x} \in \mathbb{R}^{H \times W \times 3}$ represent an RGB image of a static scene containing a *human* and an *object*. We utilize a pixel-wise corresponding control signal $\boldsymbol{c} \in \mathbb{R}^{T \times H \times W \times 3}$ to control the human's motion. The task of the model is to predict the dynamics of the object resulting from this interaction. Our objective is to realistically synthesize how the object moves or changes in response to the interaction, maintaining physical and visual consistency with the input image.

To emphasize how difficult our proposed task is, it should be noted that that (1) we only provide the human motion, where the model needs to synthesize how the object moves all by itself. We do not give any control signal regarding the *object motion*; we provide not even a bounding box around the object of interest, as in [40, 70], or sparse trajectories describing its movement, as in [73], (2) the task consists of synthesizing an entire video, including the scene, lighting changes, the human movement, object movement and camera movement: we perform no inpainting, use no UV map of the hand or object and do not copy the background from the ground truth, as in [30]. For all of this we rely on SVD as a generative prior, and on ControlNet to encode the human control into the latent space of SVD in a manner that creates controlled spatially and temporally consistent generations.

We simplify our new task by assuming that (1) the *human* and the *control signal* are always image aligned, so that the model does not have to learn complicated spatial relationships between the control signal and the human movement on top of inferring the object dynamics, (2) the control signal has pixel correspondence, meaning that there is a one-to-one mapping between pixels in the control signal and pixels in the input image, (3) the first frame of the control signal $\boldsymbol{c} \in \mathbb{R}^{t=1 \times H \times W \times 3}$ is always temporally synchronized with the input image $\boldsymbol{x} \in \mathbb{R}^{H \times W \times 3}$, and (4) the camera position stays constant over time so that the model does not have to differentiate between movement within the scene and movement of the camera. As denoted in Section 3, we cannot properly enforce SVD to keep the camera still without setting the motion ID to zero, which also does not allow for motion within the scene. Therefore, we rely on a dataset with a static camera and the control branch learning this property from the data.

As mentioned in Section 1, to the best of the author's knowledge, this research represents the first attempt to address this specific task.

4.2 Control signal

Our goal is to have an informative, 3D-aware control signal with which the human can interact with the objects in the scene, as represented by the input image. There are several design choices that can be made here, based on the dataset available, the type of control mechanism, and the amount of information that we want the network to receive. We describe in the following the alternative design choices that we have explored within the context of this thesis, along with their advantages and disadvantages. Please refer to Figure 11 for an overview of the different control signals we have considered, and the ones that we eventually chose to experiment with.

Global vs. local. The first design choice we need to address is whether the control signal will be of *global* or *local* nature. In this scenario an obvious global control signal would for instance be the rotation and translation matrices of the hand overtime. Instead, following ControlNet that employs a U-Net encoder, we opt to use a pixel-aligned aligned control signal. Such a mask not only provides the model with the location of the hand, but also provides the model with an indication on the shape of the hand.

Sparse vs. dense correspondence. Within a pixel-aligned control signal there are several alternatives, ranging from sparse information such as bounding boxes or keypoints, to more dense information such as the hand skeleton or a full hand mask. In this case, we need to keep two things in mind: the explicitness that the control signal gives at inference time, and the dataset availability. For instance, datasets with annotated bounding boxes for human body movement or hand movement are plentiful, however, a bounding box at inference time does not allow to control exactly how a person will grasp an object at the finger-level. On the other hand, if we would like to train with a more fine-grained control signal such as a highly accurate 2D mesh-based rendering of the hand position, we would be limited to datasets that provide some form of mesh for the hand, as well as its rotation and translation at each frame. Such datasets are typically much more limited in size. Within this thesis we are interested in explicit control over the human in the scene, which is why we opt for a dense correspondence control signal. We experiment with both a skeleton representation as well as a hand mask.

Ground truth vs. estimated. For both a sparse or dense correspondence, we can either use video datasets with ground truth annotations, or collect a large video dataset ourselves and use a pre-trained method to estimate these control signals. The difference here lies in the accuracy of the resulting control signal. We hypothesize that if the dataset is large enough, the accuracy of the control signal matters less since such large capable models (e.g. Stable Video Diffusion) will be able to generalize to ignore outlier control signals. Since our task is new and best practices have not yet been established, we opt to use as accurate of a ground truth control signal as we can get. This allows to perform controlled studies, where the quality of the data is not a factor. Like mentioned, this immediately also limits the datasets that we can use to conduct our experiments, diminishing diversity in terms of objects, subjects and scenes. This is a design choice within the motto of this master thesis to provide a minimal viable solution.

3D information. The amount of 3D information we provide in the control signal *could* heavily influence the results. For instance, a rendered white hand mask provides limited information on the 3D orientation of the hand over time, while a 3D volume of the hand over time would provide an incredibly rich control signal. In this project, we aim to use ControlNet, which out-of-the-box works with a 2D control signal. Since we already opted to use a dataset in which we have access to ground-truth 3D information, we are able to provide a pixel-wise correspondence map to create a pseudo-3D representation. In essence, we create a detailed map that assigns a color to each point on the 3D model of the hand or to each keypoint. We then render this colored 3D model into a 2D image as if viewed from the camera's perspective. Please refer to Section 4.4 for a detailed walk-through of our data processing pipeline.

Figure 11: Overview of control signals. Illustration of the diverse control signals we considered for the control signal. It highlights the progression from sparse to dense information.

4.3 Architecture & Training

In addressing the task of human-object interaction synthesis from a static image, this thesis proposes an integration of ControlNet with Stable Video Diffusion (SVD). This architecture is designed to control both spatial and temporal aspects of video generation, enabling the precise manipulation of a human subject interacting with an object within a scene. ControlNet introduces a control encoder that operates alongside the SVD's encoding and decoding mechanisms, connected through zero convolutional layers. This setup allows the model to incorporate specific movement instructions via control signals into the video generation process, thus achieving precise control over the depicted actions.

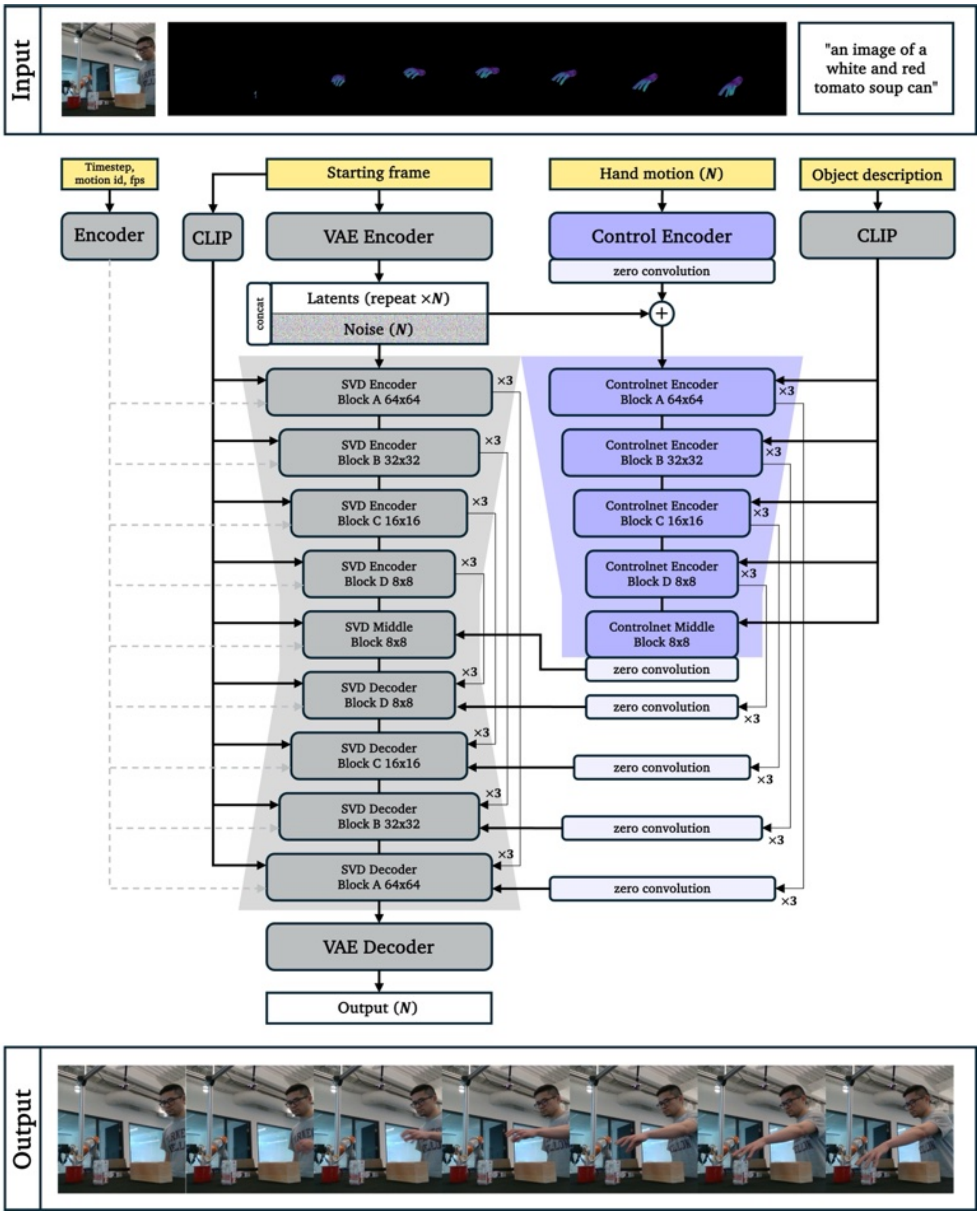


Figure 12: Architecture. The architecture receives two primary inputs: a starting frame and a detailed hand motion control signal. The starting frame serves as the visual foundation for video generation, while the hand motion control signal dictates the movements of the human subject within the video. Following ControlNet's implementation (see Section 3) the additional encoder processes the hand motion control signal, and uses the SVD backbone to generate a video that aligns with this signal through skip connections modulated by zero convolutional layers.

Our proposed architecture is shown in Figure 12. ControlNet is integrated into the SVD architecture through multiple encoder blocks, with each block of the SVD encoder paralleled by a corresponding ControlNet block. This structure allows the diffusion process to be influenced at multiple levels of abstraction. The zero convolution layers within the ControlNet blocks start with zero weights, ensuring that the initial control signals do not disrupt the pre-trained stability of SVD. As training progresses, these layers adaptively increase their influence to more significantly integrate the control signal into the generation process. The result is a sequence of frames depicting the human interacting with the object according to the controlled movements dictated by the hand motion signals. We initialize our model with the pre-trained Stable Video Diffusion (SVD) weights, specifically either the "stable-video-diffusion-img2vid"[1] or the extended "stable-video-diffusion-img2vid-xt"[2] variant. To adapt the model to our specific task, we add a ControlNet module following detailed implementation guidelines of [77]. Please refer to Section 3 for a detailed explanation of the SVD [4] and ControlNet [77] architectures. We provide a detailed overview of each step within a single training iteration in Table 2, and we refer to Table 9 in the Appendix for a detailed overview of the training-specific hyperparameters. In the following we highlight the training implementations details that are different from SVD or ControlNet, or might not be as intuitive for someone new to the field.

The SVD version that the authors provide access to is fine-tuned to a large resolution of 576×1024 in the last training stage (see Table 1), which made the authors shift the noise distribution towards slightly more noise in this stage: $\log \sigma \sim \mathcal{N}(1.0, 1.6^2)$. Since we use a smaller resolution, 512×512, we shift our noise distribution back to the noise distribution of training stage 5: $\log \sigma \sim \mathcal{N}(0.7, 1.6^2)$. We use the same learning rate with which SVD was trained: $3e-5$. One advantage of *latent* diffusion models is that during training we do not have to convert the model prediction back to the image space, since we can compute the loss within latent space. Only at inference time we use the VAE decoder to convert the latent back to image space. During training we choose to fix the FPS to 7, which falls within the range that is recommended by the authors of SVD and the authors of TRF [23]. We set the motion ID to 40 based on empirical analysis that shows this correlates to movements within the scene rather than a moving camera, please refer to Section 3.2.

While ControlNet is relatively memory-efficient compared to training the entire SVD model, we still run into memory constraints due to the mere size of the entire architecture. We use a combination of techniques to lower our memory footprint during training, including: mixed precision [42], gradient checkpointing [14], gradient accumulation steps and training on videos of 14 frames (note, this is the native context length of the "stable-video-diffusion-img2vid" version but not for the "stable-video-diffusion-img2vid-xt" version, which was trained with an identical architecture but on a context length of 25). Using these techniques, we just barely fit a batch size of 2 (28 frames) on one NVIDIA H100 80GB GPU. For training, we use 4 NVIDIA H100 80GB GPUs in a multi-GPU training setup and 2 gradient accumulation steps, giving us an effective batch size of 16. In order to stabilize the gradient flow of this relatively small batch size, we also use gradient clipping with the max grad norm setting. We noticed that this leads to considerable more stable training.

[1] `https://huggingface.co/stabilityai/stable-video-diffusion-img2vid`

[2] `https://huggingface.co/stabilityai/stable-video-diffusion-img2vid-xt`

Step	Description
1	Sample a video from the dataset: $\mathbf{z} \in \mathbb{R}^{14\times3\times512\times512}$ together with its accompanying control signal: $\mathbf{c} \in \mathbb{R}^{14\times3\times512\times512}$.
2	Retrieve the first frame of the video: $\mathbf{f} \in \mathbb{R}^{3\times512\times512}$, sample augmentation noise $\mathbf{n} \sim \mathcal{N}(0,1)$, sample a noise augmentation strength $\sigma_{\text{aug}} \sim \mathcal{N}(-3.0, 0.5^2)$, scale the augmentation noise with this strength, and add the noise to the first frame: $\mathbf{f}' \leftarrow \mathbf{f} + \sigma_{\text{aug}} \cdot \mathbf{n}$ The augmentation noise is added to make the model robust to different image qualities at inference time.
3	Convert the video and the augmented first frame to the latent space of the `VAE` using mean sampling: $\mathbf{z}_{\text{latent}} \leftarrow \texttt{VAE}(\mathbf{z})$ and $\mathbf{f}'_{\text{latent}} \leftarrow \texttt{VAE}(\mathbf{f}')$.
4	Sample noise $\mathbf{n} \sim \mathcal{N}(0,1)$, a noise level $\log\sigma \sim \mathcal{N}(0.7, 1.6^2)$, and add the noise following: $\mathbf{z}'_{\text{latent}} \leftarrow \mathbf{z}_{\text{latent}} + \log\sigma \cdot \mathbf{n}$. Then scale the noisy latent using the $c_{\text{in}}(\sigma)$ pre-conditioning function: $\mathbf{z}'_{\text{latent\_scaled}} \leftarrow \mathbf{z}'_{\text{latent}} \cdot c_{\text{in}}(\sigma)$. This pre-conditioning function helps to bring the noisy latent within an appropriate scale in which neural networks are effective, based on the sampled noise level. We also map noise level σ to a conditioning input using the $c_{\text{noise}}(\sigma)$ preconditioning function, and later provide this conditioning input to the U-Net and ControlNet for noise level conditioning.
5	Convert the noise augmented first frame to a CLIP embedding: $\mathbf{f}'_{\text{CLIP}} \leftarrow \texttt{CLIP}_{\text{image}}(\mathbf{f}')$.
6	Convert the FPS, motion ID and the sampled augmentation noise strength σ_{aug} to one embedding that we later feed into the cross-attention layers of the denoising U-Net and ControlNet: $\mathit{ids} \leftarrow [\text{FPS}, \text{motion ID}, \sigma_{\text{aug}}]$.
7	Repeat the first frame: $\mathbf{f}'_{\text{latent}} \in \mathbb{R}^{14\times3\times512\times512} \leftarrow \mathbf{f}'_{\text{latent}} \in \mathbb{R}^{3\times512\times512}$ and concatenate it to the noisy latent: $\mathbf{zf}'_{\text{latent}} \in \mathbb{R}^{2\times14\times3\times512\times512} \leftarrow \mathbf{z}'_{\text{latent\_scaled}} \oplus \mathbf{f}`_{\text{latent}}$.
8	Convert the control signal to latent space dimensions using a tiny encoder: $\mathbf{c}_{\text{latent}} \leftarrow \mathcal{E}(\mathbf{c})$ and perform the `ControlNet` forward pass: $\mathbf{h}_{\text{ControlNet}} \leftarrow \texttt{ControlNet}(\mathbf{zf}'_{\text{latent}}, \mathbf{c}_{\text{latent}}, \mathbf{f}'_{\text{CLIP}}, \mathit{ids}, c_{\text{noise}}(\sigma))$.
9	$\text{pred}_{\text{UNet}} \leftarrow \texttt{UNet}(\mathbf{zf}'_{\text{latent}}, \mathbf{h}_{\text{ControlNet}}, \mathbf{f}'_{\text{CLIP}}, \mathit{ids}, c_{\text{noise}}(\sigma))$.
10	Retrieve an estimation of $\mathbf{z}_{\text{pred}}$ through the $c_{\text{out}}(\sigma)$ and $c_{\text{skip}}(\sigma)$ preconditioning functions : $\mathbf{z}_{\text{pred}} \leftarrow \text{pred}_{\text{UNet}} \cdot c_{\text{out}}(\sigma) + c_{\text{skip}}(\sigma) \cdot \mathbf{z}'_{\text{latent}}$. Here $c_{\text{out}}(\sigma)$ scales the output of the `UNet` back to the range of the noise level and $c_{\text{skip}}(\sigma)$ modulates how much of the original input we retrain.
11	Compute a weighted MSE loss between the prediction and target latents: $\mathcal{L} = \mathbb{E}_{\sigma}\left[\lambda(\sigma)\ \lVert\mathbf{z}_{\text{pred}} - \mathbf{z}_{\text{latent}}\rVert_2^2\right]$.
12	Perform backpropagation with respect to the weights of `ControlNet`.

Table 2: Training iteration. Order of one training iteration of our proposed architecture. The iteration is shown for a batch size of one to not clutter the notation unnecessarily.

Originally, the base SVD model is a text-to-image model, conditioned on the CLIP embedding version of a textual description of the video. Only later, in training stage 5 and 6 of SVD, is the embedding changed to the CLIP embedding of the first image, see Section 3.2. CLIP is a joint image and text embedding model that maps both text and images to the same embedding space, meaning that an image of a banana and the sentence “an image of a banana” will be encoded with very similar embeddings. The CLIP features of the conditioning provide high-level global features about the conditioning image, but objects often do not dominate the conditioning image, hence they would be highlighted less in the CLIP features of the conditioning image. Therefore, we experiment using this joint embedding property of CLIP to inject additional information about the object to the ControlNet encoder, through a textual description of the object to be grasped. Following the text processing in CLIP, we prepend the name of the object with “an image of” and encode each object in the following manner:

$$\mathbf{o}'_{\text{banana\_CLIP}} \leftarrow \texttt{CLIP}_{\text{text}}(\text{“an image of a yellow banana”}) \tag{16}$$

this step is performed after step 5 of our training procedure, see Table 2. Then we adjust training step 8 of our training procedure to:

$$\mathbf{h}_{\text{ControlNet}} \leftarrow \texttt{ControlNet}(\mathbf{zf}'_{\text{latent}}, \mathbf{c}_{\text{latent}}, \mathbf{o}'_{\text{banana\_CLIP}}, ids, c_{\text{noise}}(\sigma)) \tag{17}$$

Even though the text and image are naturally embedded within the same CLIP space, since the weights of SVD remain frozen, SVD expect high level semantics of the scene rather than high level semantic information on the object. Therefore, we only supply ControlNet with the object-specific CLIP embedding and supply the CLIP embedding of the conditioning frame to the SVD encoder-decoder pipeline. In this manner, we hope to enrich the conditioning information ControlNet can use to ensure object consistency and differentiate between objects that partially occlude each other. Please refer to Appendix B for a list with each object description.

4.4 Dataset

In this master thesis, we opted to use DexYCB [11] for our experiments and model selection. DexYCB serves as a sophisticated benchmark for analyzing hand-object interactions with a specific focus on grasping everyday objects. Specifically, DexYCB captures full grasping processes, from hand approaching, opening fingers, contact, to holding the object stably. The dataset has been pivotal for developing and evaluating algorithms in the domains of 3D hand pose estimation, 3D object pose estimation and the interaction dynamics between hands and objects.

The data is captured synchronously from 8 distinct viewpoints using a multi-camera setup, at a resolution of 640×480 pixels at 30 frames per second. Each video sequence approximately lasts for 3 seconds, ensuring each action of grasping is captured from start to finish. Ten subjects were involved in the data collection process, each engaging with a predefined set of household and tool-related objects. These include items such as food cans, tools, and various everyday objects, designed to represent a range of sizes and complexities in terms of shape and required handling, see Figure 15. In addition, the dataset provides the coefficients for the 3D hand parametric model of [53], including pose and shape information. The objects are represented using a standard 3D pose model, referring to rotation and translation. While DexYCB is robust in many respects, it does come with limitations such as motion blur, see Figure 14, which can affect the quality of the videos and the accuracy of pose estimation.

Attribute	**Specification**
Number of objects	20
Number of subjects	10
Number of views	8
Number of sequences	1000
Number of videos	8000
Video length (seconds)	∼3
Frame rate (FPS)	30
Resolution (pixels)	640×480
Annotation method	Crowd sourced
Keypoints per object	2
Keypoints per hand	21
Hand model	MANO [53]
Pose estimation method	Optimization

Figure 13: DexYCB. Specification of attributes.

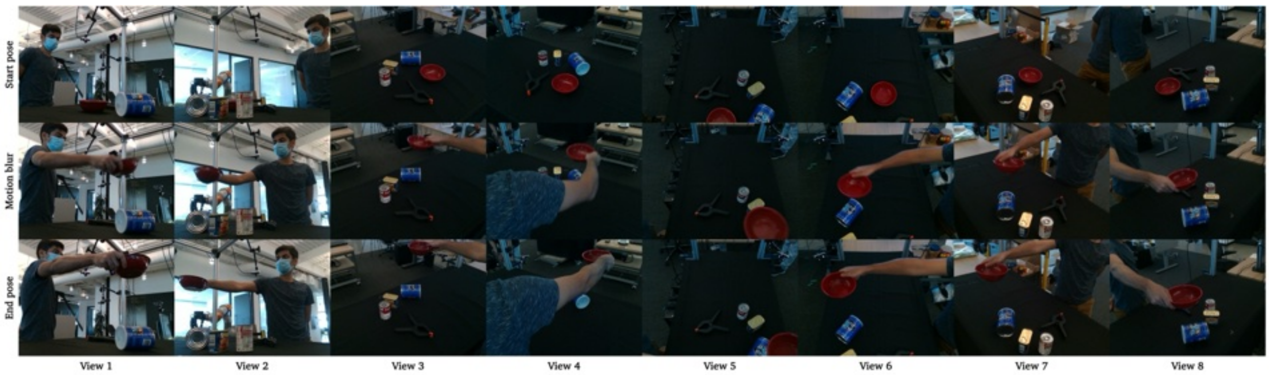


Figure 14: DexYCB Views. The eight different views in the DexYCB dataset. (a) Starting position (b) Motion blur (c) End position.

Please refer to Figure 14 for an impression of the different views in the DexYCB dataset, including the motion blur that the dataset contains and a common starting and end position of the human subject in one sequence. These samples are from the original DexYCB dataset before our data processing pipeline is applied. Notice that some views have a primarily dark background (view 3-8), while others are depicting a more in-the-wild type of setting (view 1-2). Please refer to Figure 15 for an impression of all objects in the DexYCB datasets.

Figure 15: DexYCB Objects. From left to right, top to bottom: Banana; Bleach cleanser; Bowl; Cracker box; Extra large clamp; Foam brick; Gelatin box; Large marker; Master chef can; Mug; Mustard bottle; Potted meat can; Scissors; Tomato soup can; Tuna fish can; Wood block; Pitcher base; Power drill; Pudding box; Sugar box.

Processing

To represent 3D hand pose, DexYCB uses MANO [53], which is a well-established 3D parametric human hand model. MANO defines a hand model that takes as input pose $\theta \in \mathbb{R}^{48}$ and shape $\beta \in \mathbb{R}^{10}$ parameters, and returns a 3D hand mesh $V \in \mathbb{R}^{778\times 3}$ and 3D joint keypoints $V \in \mathbb{R}^{21\times 3}$. The shape parameters define the static, anatomical structure of the hand. They capture variations in hand size, bone lengths, and proportions that are specific to an individual. This means that the shape parameters are crucial for modeling the unique geometric characteristics of a person's hand, which do not change as the hand moves. The pose parameters define the dynamic aspects of the hand, specifically how the joints of the hand are articulated. They are essential for capturing the wide ranges of motion of the hand, from different finger configurations to the overall orientation.

In DexYCB, the pose parameters are estimated in each frame by minimizing a loss function between annotated hand keypoints and the MANO keypoints, whereas the shape parameters are estimated once for each subject and kept fixed. For each frame, DexYCB supplies these shape and pose parameters in addition to the global rotation and translation of the hand, which we use to generate the control signal in the form of 3D joint keypoints or a hand mesh, aligned with hand in each image frame. We apply two distinct colormaps to the rendered keypoints or mesh, one for each hand. Please refer to Figure 16 for an example of the applied colormap. These pixel-wise correspondence maps ensure that our training pipeline does not have to focus on learning accurate hand shape consistency, and that the generated video's hand shape accurately matches the reference image while following the dynamic poses from the driving videos. We render the conditioning signal with a resolution of 640×480, the same resolution as the videos supplied in DexYCB. We noticed that most of the human-object interaction happens within the center of the frames, which is why we center-crop both the video and the conditioning signal to 480×480 and resize them to 512×512. In this way we strike a balance between memory constraints, image size and the area of the video data that has the most human-object interaction.

For our training scenario, we would like to provide a pixel-wise correspondence map as conditioning signal, which means that for each starting frame that we sample we would like the hand to be visible within the frame. DexYCB provides the rotation, translation and DexYCB parameters for each sequence irrespective of whether the hand is visible from all views of that sequence, which means that for some videos the hand is not visible in the starting frames while we do have a control signal. This effect is amplified by the way we center-crop the data. Therefore, for each individual video we remove starting frames where the conditioning signal is not within frame, to ensure that we always have a starting frame with an accompanying control signal. For training, we would like to stay around video data at around 7 frames per second. Since DexYCB provides videos at 30 frames per second, we sub-sample frames with a step size of 4, effectively giving us video data at 7.5 frames per second. This means that in order for a DexYCB video to be used in our training data, it should be at least $14 \times 3 = 53$ frames long. We remove videos that are shorter than this length and videos that have intermediate missing hand registrations, which removes 1240 out of 8000 videos.

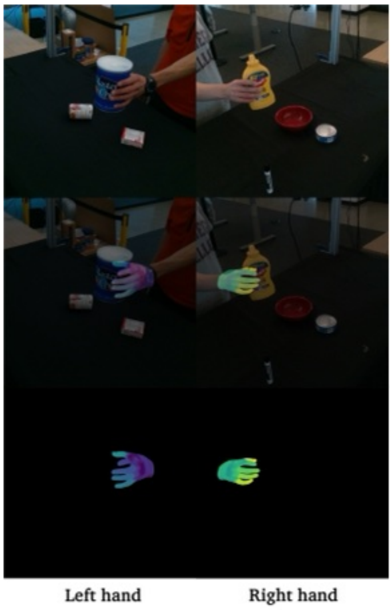


Figure 16: MANO pixel-wise correspondence. The left and right hand are rendered using two non-overlapping colormaps.

Data split

We split the DexYCB dataset object-wise, as detailed in Table 3. In the exploratory stage of this project, we noticed great variability in the performance of the model for individual object classes, often correlating with high-frequency details and unique shapes. We argue that some objects in DexYCB are inherently harder to generatively model due to their complex shapes, textures, or interaction dynamics. Take the power drill for instance. This object has quite a complex 3D shape, is sometimes partially occluded in the starting frame and the black parts of the power drill are sometimes indistinguishable from the primarily dark background of views 3, 4, 5 and 6, as can be seen in Figure 14. This makes some objects inherently more challenging to detect, harder to model generatively, or both.

Therefore, we argue that for a traditional train, validation and test split for as small of a dataset as DexYCB, the validation would not adequately represent the diversity of objects, limiting model selection. For instance, if a relatively easy object such as the marker were to be in the validation set, we would risk drawing incorrect conclusions about our model's overall performance. Rather than evaluating whether our model can generalize to new objects, we would in practice be evaluating whether the model can transfer its learnings on relatively harder objects, to simpler objects.

To mitigate this, we had two options: (1) decreasing the number of objects in the training set or (2) merging the test set into validation. We opted for the latter, since a sufficient amount of training data is essential for the model to learn effectively and generalize to new manipulation scenarios. Therefore, we merge the testing set into validation, creating one larger validation set comprising 4 objects. This enlarged set better captures the range of object difficulties within DexYCB and provides a more reliable estimate of the model's generalization capabilities.

We omit 10 random sequences from the training set for each object to create an in-distribution validation set with unseen sequences. Please refer to Figure 17 for an overview of the dataset distribution over views, subjects and objects for the training and OOD validation sequences splits.

Split	**Type**	**# of Sequences**	**Objects**
Train	ID	5204	Banana; Bleach cleanser; Bowl; Cracker box; Extra large clamp; Foam brick; Gelatin box; Large marker; Master chef can; Mug; Mustard bottle; Potted meat can; Scissors; Tomato soup can; Tuna fish can; Wood block;
Validation	ID	160	*Same as training objects (unseen sequences)*
Validation	OOD	1396	Pitcher base; Power drill; Pudding box; Sugar box

Table 3: Data split. Training: 16 objects and Validation (OOD): 4 objects.

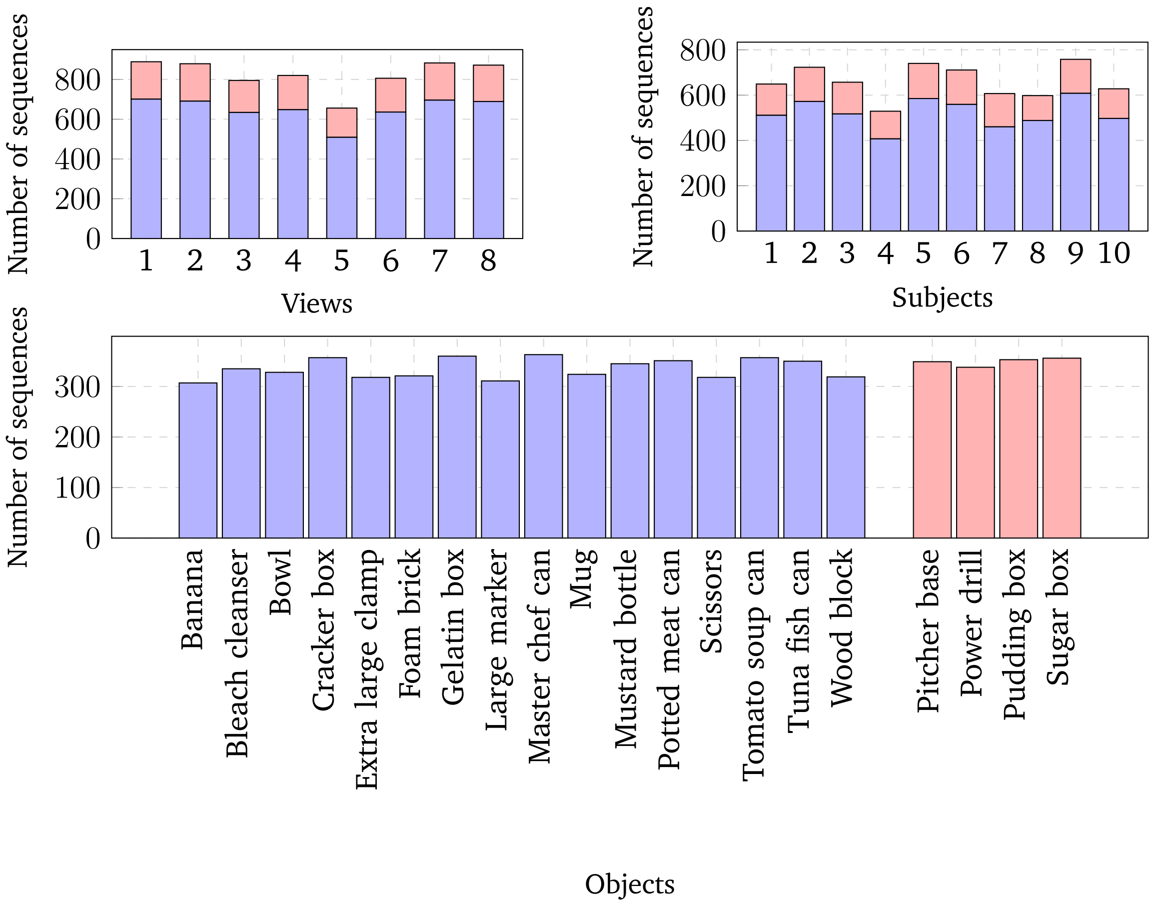


Figure 17: Dataset characteristics. The dataset is balanced over views, subjects and objects. Training sequences in blue and OOD validation sequences in red. ID validation sequences not shown for figure clarity.

5 Results

In this section, we present the results of our experiments on human-object interaction (HOI) synthesis. We first describe the different model variants we explored, followed by the evaluation metrics used to assess their performance. We then present the results of our experiments, both quantitatively and qualitatively.

5.1 Model variants

We explore several variations of our model architecture and training strategies, aiming to understand their impact on video generation performance.

Firstly, we consider two versions of the Stable Video Diffusion (SVD) model, one fine-tuned on a shorter 14-frame context length and another on a longer 25-frame context length. This investigation examines the effect of context length on the model's ability to capture temporal dynamics and object consistency. Secondly, we experiment with two types of control signals: skeleton-based (a sparse representation of hand pose) and mesh-based (a dense representation providing detailed shape and pose information). This comparison aims to assess the impact of control signal detail on hand movement realism and object interaction accuracy. Thirdly, we evaluate the effect of conditioning the ControlNet on either the CLIP embedding of the first frame or a textual description of the object, hypothesizing that the latter could enhance object understanding and dynamic generation. Lastly, we manipulate the information flow to the ControlNet, comparing a variant receiving both the initial frame's visual information and control signal, with a variant receiving only the control signal.

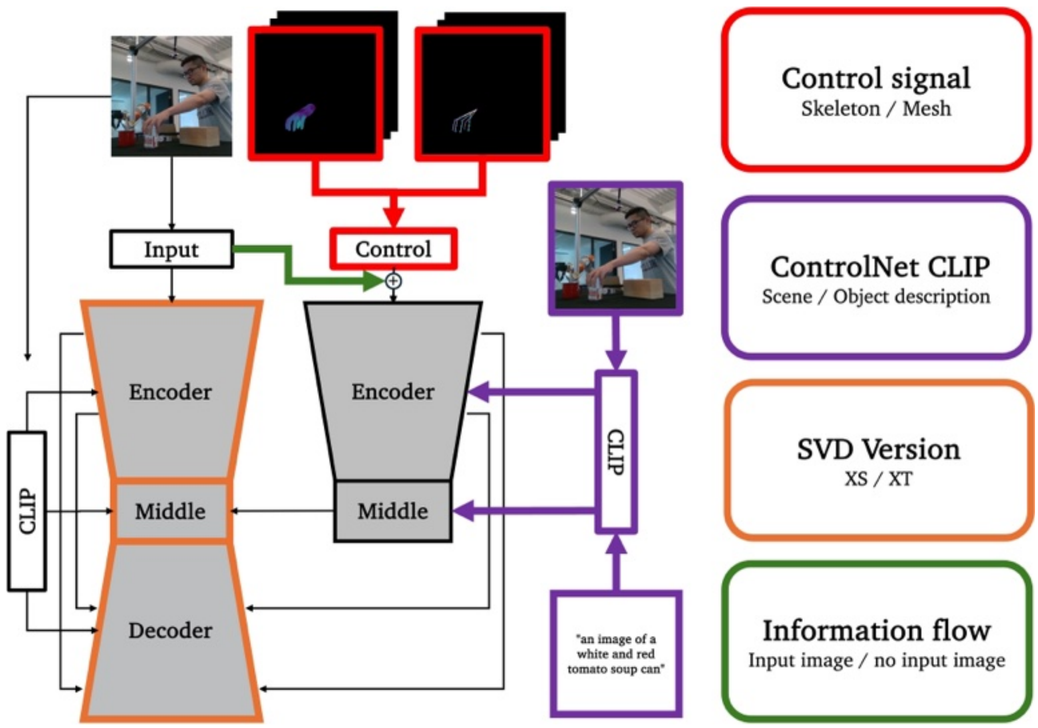


Figure 18: Model variants. We explore four variations of our framework, focusing on: (a) skeleton-based vs. mesh-based control signals, (b) conditioning ControlNet with the CLIP embedding of the first frame, or the CLIP embedding of a textual object description, (c) SVD originally fine-tuned on a 14-frame context length vs. SVD originally fine-tuned on a 25-frame context length, and (d): manipulating the information flow to the ControlNet, either providing both the input image and control signal, or only the control signal.

In general, we expect that reducing information might lead to a more generalized control strategy. We hypothesize that the model variants with richer information (longer context length, mesh-based control signals, and additional CLIP embeddings) will outperform simpler variants on in-distribution data due to better object understanding and temporal coherence. However, we also expect that these variants might be more prone to overfitting and may not generalize as well to out-of-distribution data compared to variants with reduced information flow. Please refer to Figure 18 for a visual overview of these variants.

5.2 Evaluation metrics

To evaluate the generated videos, we employ various metrics designed to capture different aspects of video quality, with an emphasis on perceptual similarity and temporal coherence. Our primary metric is Fréchet Video Distance (FVD) [63], complemented by several additional metrics. *Importantly*, even though our model predicts 14 frames, the first frame of our generated video is in essence a reconstruction of the conditioning image. We noticed that the metrics for the first frame were out-of-the-ordinary good, while in theory the model could simply reconstruct the conditional latent of the conditioning frame. Therefore, in order to avoid inflating our metrics and to maintain scientific rigor, we exclude the first frame from both the ground truth and the generated videos. We report the evaluation metrics for our training split, in-distribution validation split, and out-of-distribution validation split (see Section 4.4).

FVD. FVD is a perceptual metric that assesses the similarity between two distributions of videos, inspired by the Fréchet Inception Distance (FID) [27] used for images. It leverages features extracted from a pre-trained Inception 3D ConvNet (I3D) [9] to encapsulate both spatial and temporal information within videos. This enables FVD to quantify the overall perceptual similarity between the distribution of real videos and the distribution of generated videos. Our In-Distribution (ID) evaluation set comprises 160 sequences, which might lead to slightly inflated absolute FVD values due to the limited sample size. This is because, FVD compares the mean and covariance of the feature embedding of the ground truth and generated videos. With a smaller sample size, this estimate becomes less accurate and more prone to variability, leading to higher FVD values even if the quality of the videos remain the same. For the computation of the FVD score for the in-distribution and out-of-distribution data we use the full data split of 160 and 1396 videos respectively, whereas for the FVD score on our training data, we randomly sample 1040 videos from our training split. This random subset is sampled once and fixed for all experiments.

Image quality metrics. To gain a more comprehensive understanding of the image quality of our video generations, we also report complementary metrics. First of all, we report the Mean Squared Error (MSE), which measures the average squared difference between pixel values in corresponding frames of the real and generated videos, providing an indication of pixel-level accuracy. Importantly, we report the MSE for the *object area*. Since we have access to the 6D object pose for each frame, we can render out a binary mask which indicates the object area in the frame. Importantly, we render out this mask with hand occlusions, which means that when the hand is positioned between the object and the camera, this part is omitted from the object mask. Then we use this mask to compute the MSE score for the object area, between the ground truth and generated video. Secondly, we report the Learned Perceptual Image Patch Similarity (LPIPS) [78], which evaluates the perceptual similarity between images, designed to reflect a score closely related to human visual perception. Thirdly, we report the Peak Signal-to-Noise Ratio (PSNR), which quantifies the ratio between the maximum possible power of a signal and the power of corrupting noise, indicating the level of noise or distortion present in the generated videos. Often this metric does not necessarily align with the human perception of image quality.

Lastly, we report the Structural Similarity Index Measure (SSIM) [72], which encapsulates the structural similarity between images based on luminance, contrast, and structure. This metric helps gauge the preservation of structural details in the generated videos.

While these additional metrics offer valuable insights into specific aspects of video quality, FVD remains our primary metric due to its holistic approach. By encapsulating both visual quality and temporal coherence, FVD aligns closely with human perception of video quality [63].

5.3 Evaluation settings

Due to the highly probabilistic and generative nature of diffusion models, we perform inference with three different random seeds for each variant on which we report our evaluation metrics, to ensure robustness and assess variability in the results. After having ran inference, we first compute a per-sample mean and standard deviation, which we then average over the data split. Please refer to Table 4 for an overview of the hyperparameters used for inference in our experiments. These settings are consistent across all evaluations, ensuring fair and comparable comparisons between different model variants and conditioning variants.

Importantly, we would like to emphasize that *all* hyperparameters, such as batch size, training iterations, motion ID, frames per second, data split, and the random seed were held constant across experiments. This consistency makes it more likely that any observed differences in validation performance are attributable to variations in the model version, model architecture, control signals and conditioning that we experiment with.

Category	**Hyperparameter**	**Setting**
General	Number of inference steps	25
	Per-GPU batch size	4
	Number of frames	14
	Noise augmentation strength	0.02
	Seeds	[42, 43, 44]
Specific	Min guidance scale	1.0
	Max guidance scale	3.0
	ControlNet condition scale	1.0
	Motion ID	40
	FPS	7

Table 4: Inference hyperparameters

5.4 Variant evaluation

SVD Version. As mentioned before, Blattmann et al. [4] released two versions of the SVD image-to-video model: one fine-tuned on a context length of 14 frames and another one on 25 frames. Here the context length refers to the number of frames the model considers simultaneously during *training*, influencing its ability to capture longer-range temporal relationships. While both versions share the same underlying architecture and can technically handle any context length at inference time (within GPU memory limits), the model fine-tuned on 25 frames has temporal layers specifically optimized to capture longer-range temporal relationships. This optimization could potentially enhance the model's ability to maintain object consistency over time. Due to memory constraints, we fine-tuned our architecture with a context length of 14 frames for both SVD model versions. We hypothesize that since the longer context length model has been optimized for longer-range temporal relationships, it will maintain better object consistency and overall temporal coherence in the generated videos, even when generating sequences shorter than its training context length.

Split	Control	FVD ↓		MSE ↓		LPIPS ↓		PSNR ↑		SSIM ↑	
		mean	std	mean	std	mean	std	mean	std	mean	std
Train	Small	34.509	0.716	0.058	0.012	0.111	0.009	26.827	0.848	0.878	0.007
	Large	40.898	1.507	0.070	0.016	0.125	0.011	25.979	0.861	0.870	0.008
Val ID	Small	122.255	1.386	0.086	0.014	0.127	0.007	25.202	0.634	0.867	0.005
	Large	133.876	5.458	0.087	0.015	0.135	0.009	24.902	0.697	0.862	0.006
Val OOD	Small	**62.178**	2.459	**0.099**	0.034	**0.142**	0.026	**24.616**	1.525	**0.860**	0.018
	Large	63.609	1.181	0.101	0.034	0.151	0.026	24.288	1.473	0.855	0.018

Table 5: SVD version performance. In this table we show the performance of two different SVD versions. One is originally finetuned on a 14-frame context-length and one originally finetuned on a 25-frame context-length. Both are evaluated on a 14-frame context-length.

The results of this experiment are shown in Table 5. *Contrary* to our expectation, the 14-frame context length version slightly outperformed the 25-frame version across all evaluation metrics, including the most important one: FVD. This was surprising because we anticipated that the 25-frame version, with its enhanced temporal layers, would excel at maintaining consistency across the generated 14-frame videos. We believe that this unexpected result could be attributed to the MSE training loss function used by SVD, which is computed by taking the mean of the frame-wise MSE. Therefore, this loss function prioritizes overall average performance over longer context lengths but does not explicitly ensure better quality in the initial frames. Hence, since the shorter context length model is especially fine-tuned to perform better on 14 frames on average, this shorter context-length version of SVD outperforms the longer context-length version.

Control signal. For our proposed task, the selection of an effective control signal representation is critical as it can provide information on how an object is grasped, and accordingly what its dynamics could be. As mentioned in Section 4.2, we experiment with two types of local control signals: sparse pixel-wise correspondence (skeleton-based) and dense pixel-wise correspondence (mesh-based). We train two versions of our pipeline, where the only difference is that one receives the skeleton-based control signal and the other receives the mesh-based control signal.

Split	Control	FVD ↓		MSE ↓		LPIPS ↓		PSNR ↑		SSIM ↑	
		mean	std	mean	std	mean	std	mean	std	mean	std
Train	Skeleton	39.644	0.842	0.057	0.012	0.119	0.011	26.413	0.943	0.873	0.009
	Mesh	34.509	0.716	0.058	0.012	0.111	0.009	26.827	0.848	0.878	0.007
Val ID	Skeleton	135.025	5.741	0.086	0.014	0.133	0.009	25.009	0.684	0.863	0.006
	Mesh	122.255	1.386	0.086	0.014	0.127	0.007	25.202	0.634	0.867	0.005
Val OOD	Skeleton	72.277	3.272	**0.098**	0.034	0.149	0.027	24.420	1.535	0.856	0.019
	Mesh	**62.178**	2.459	0.099	0.034	**0.142**	0.026	**24.616**	1.525	**0.860**	0.018

Table 6: Control signal performance. In this table we show the performance of two different types of control signal, a skeleton-based pixel-wise correspondence signal and a mesh-based pixel-wise correspondence signal.

The results are shown in Table 6. In general and as expected, the mesh-based control signal seems to outperform the skeleton-based control signal on all dataset splits for FVD, including on the out-of-distribution validation split. It is also worth noting that the difference in performance is more pronounced on the out-of-distribution (Val OOD) split than on the training split. We hypothesize that for out-of-distribution samples, the mesh rendering provides a clear prior on the shape of the hand and is more informative on the exact hand orientation than the skeleton. While the mesh-based control signal demonstrates a slight advantage, the marginal improvement may not justify the additional complexity and computational cost associated with acquiring and processing mesh data. Interestingly, the performance difference in other metrics is negligible. This suggests that a coarse control signal is still sufficient to accurately generate each frame's components, while the better FVD score of the mesh representation suggests that a finer control is necessary for the motion aspect of the problem.

ControlNet CLIP. In our experiments, we investigated the impact of providing the ControlNet encoder with the CLIP embedding of a textual description of the target object. Please refer to Section 4.3 for an explanation of this training strategy. We hypothesize that by conditioning the ControlNet on a CLIP embedding of a textual description of the target object, our architecture can better understand the identity and shape of the target object, leading to improved object dynamic generation.

Split	Control	FVD ↓		MSE ↓		LPIPS ↓		PSNR ↑		SSIM ↑	
		mean	std	mean	std	mean	std	mean	std	mean	std
Train	First frame	40.898	1.507	0.070	0.016	0.125	0.011	25.979	0.861	0.870	0.008
	Object description	40.595	1.056	0.070	0.015	0.124	0.010	25.963	0.860	0.870	0.008
Val ID	First frame	133.876	5.458	0.087	0.015	0.135	0.009	24.902	0.697	0.862	0.006
	Object description	129.886	4.476	0.088	0.014	0.133	0.008	24.983	0.624	0.862	0.006
Val OOD	First frame	**63.609**	1.181	**0.101**	0.034	**0.151**	0.026	**24.288**	1.473	**0.855**	0.018
	Object description	84.246	1.844	0.106	0.035	0.155	0.026	24.097	1.436	0.852	0.018

Table 7: CLIP embedding performance. In this table we show the performance of two different types of CLIP embeddings provided to the ControlNet encoder, one receives the CLIP embedding of the conditioning frame and one receives the CLIP embedding of a textual description of the object to be grasped.

As shown in Table 7, this approach yielded mixed results. While the model trained with both scene and object CLIP embeddings outperformed the baseline (only embedding of the first frame) on the training and ID validation sets, it struggled significantly on the (OOD) validation set.

The improved performance on in-distribution data suggests that the object CLIP embedding does indeed provide valuable information. Even when the object is partially occluded in the initial frame, the model can leverage the learned association between the textual description and visual features to generate an object that resembles those in the training set. This indicates that the model effectively utilizes the object's semantic information during generation. The poor performance on the OOD set highlights an important challenge when using such rich information in combination with small datasets: overfitting. By relying heavily on the object CLIP embeddings, the model learns to associate specific CLIP embeddings with specific object appearances from the training set. This leads to a lack of generalization when encountering novel objects with unseen embeddings. We hypothesize that using a CLIP embedding of the target object could be a valuable training technique, *if* your dataset is large and diverse enough so that the model learns a more general mapping between the identity and shape of an object and the CLIP embedding.

Information Flow. One key consideration in our architecture is the amount of detailed information provided to the ControlNet encoder. In our experiments, we manipulated this information flow to investigate its impact on the model's ability to learn a generalizable control signal for human-object interaction. We specifically examined the role of the initial frame's latent representation. In the baseline model, this representation is concatenated with the control signal, providing the ControlNet encoder with detailed pixel-level information about the scene, subject, and object. We hypothesize that by omitting this detailed information and providing the encoder only with the control signal and the CLIP embedding of the first frame, the model would be encouraged to learn a more generalized control signal.

Split	Control	FVD ↓		MSE ↓		LPIPS ↓		PSNR ↑		SSIM ↑	
		mean	std	mean	std	mean	std	mean	std	mean	std
Train	Input and control	40.595	1.056	0.070	0.015	0.124	0.010	25.963	0.860	0.870	0.008
	Control	47.332	1.387	0.076	0.016	0.143	0.013	24.945	0.899	0.856	0.009
Val ID	Input and control	129.886	4.476	0.088	0.014	0.133	0.008	24.983	0.624	0.862	0.006
	Control	148.001	2.699	0.089	0.016	0.148	0.008	24.220	0.672	0.851	0.007
Val OOD	Input and control	84.246	1.844	0.106	0.035	**0.155**	0.026	**24.097**	1.436	**0.852**	0.018
	Control	**78.106**	0.640	**0.104**	0.035	0.166	0.027	23.650	1.437	0.843	0.019

Table 8: Information flow performance. In this table we show the performance of two different types of information flow in our architecture. In one version, the ControlNet encoder receives both the input signal as well as the control signal and in another version the ControlNet encoder receives only the control signal

The results of this experiment are shown in Table 8. In line with our expectation, the model trained with the reduced information flow (control signal only) performed slightly worse on the training and in-distribution (ID) validation sets. The absence of detailed information on the scene, seems to make it more challenging for the model to learn specific control patterns tailored to the training data. Importantly, the model with reduced information flow does significantly outperform the baseline on the out-of-distribution (OOD) validation set. This indicates that by focusing solely on the control signal and high-level semantic information (i.e., the CLIP embedding), the model learns a more generalizable control strategy that can be applied to novel objects.

5.5 Object-wise performance

In Figure 19 we show the MSE, LPIPS, PSNR and SSIM metrics for each individual object on the Train and Validation OOD splits. We do not report the per-object FVD score, since for each object we only have 300 sequences that form 20 different sub-distributions. Computing the FVD score on such a small distribution that is also not fixed over the objects, like we have for our in-distribution validation split, would lead to very noisy and inconclusive evidence. The metrics are reported for the model variant that was trained with the mesh-based control signal, using the smaller context-length SVD version, conditioned on the CLIP embedding of the first frame and receiving both the input and the control signal. In general, we can see that objects that do not have high-frequency details and mostly are homogeneous in colour, have the best performance. These are objects such as the mug or the bowl. Objects that have a detailed texture in general perform less well, such as the cracker box or beach cleaner.

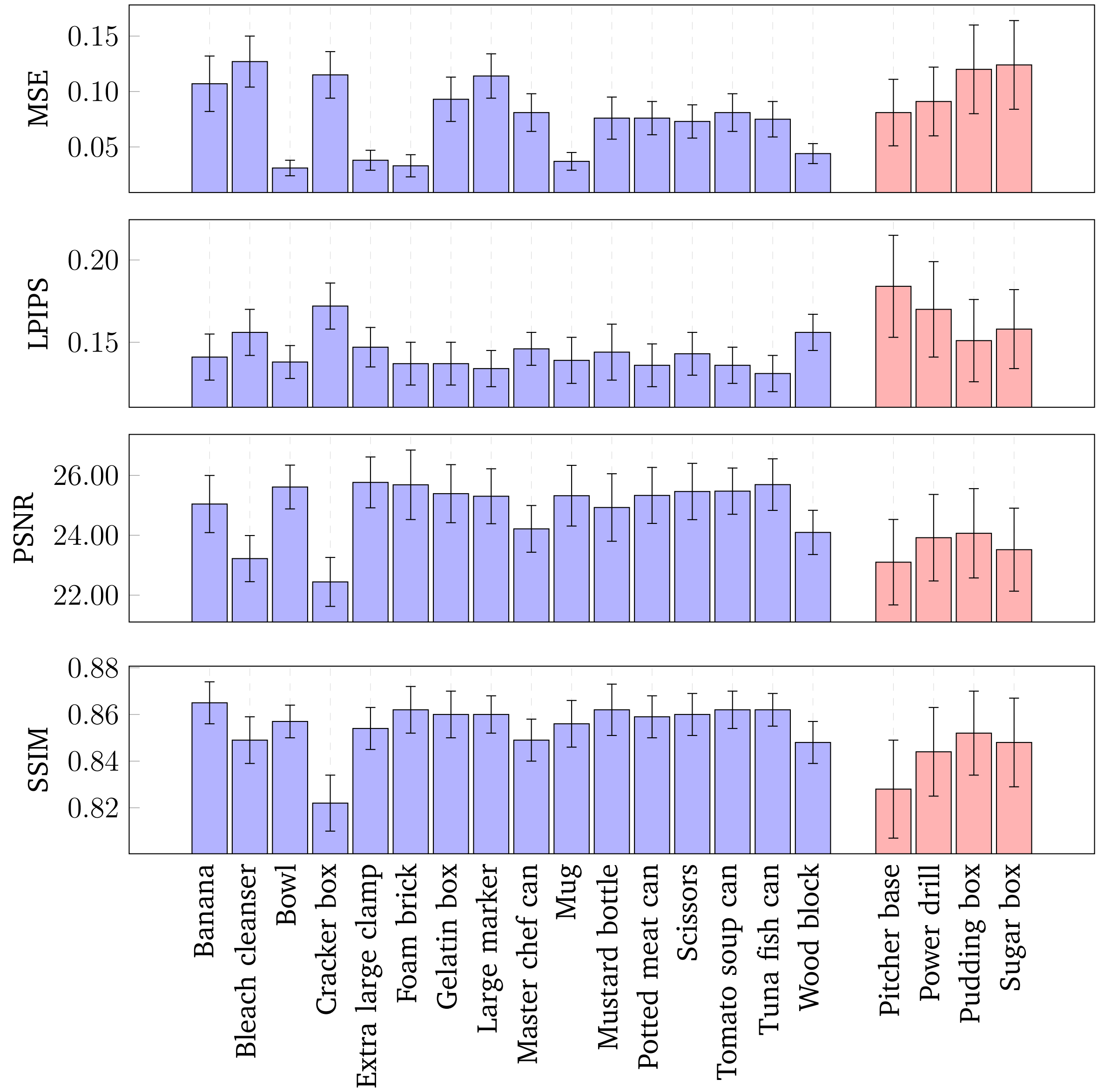


Figure 19: Object-wise performance. Quantitative evaluations for each individual object on the training set (right, blue) and validation set (left, red).

5.6 Frame-wise performance

In Figure 20 we show the MSE, LPIPS, PSNR and SSIM metrics over the course of the videos on the Validation OOD split. We do not report the first frame performance as mentioned in Section 5.2. The metrics are reported for the model variant that was trained with the mesh-based control signal, using the smaller context-length SVD version, conditioned on the CLIP embedding of the first frame and receiving both the input and the control signal. In general, we can see that the performance over the course of the generated video worsens. We see the same phenomena for vanilla SVD, similarly to other researchers who worked with this model [23]. Therefore, we find it difficult to conclude whether this worsening of performance over the course of the video is due to our control mechanism or due to how SVD was trained.

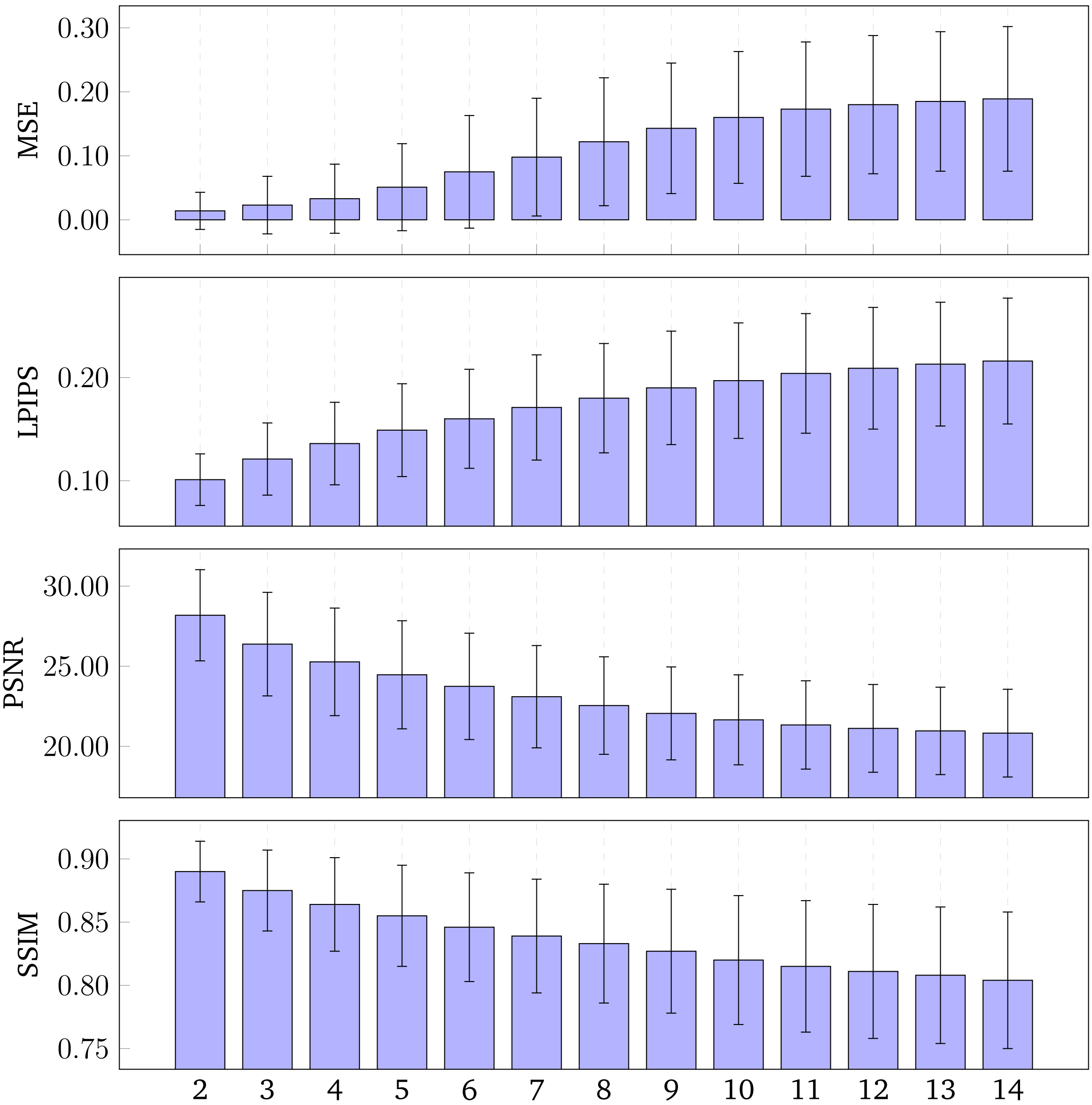


Figure 20: Frame-wise performance. Quantitative evaluations for each individual frame in the generated video.

5.7 Qualitative Results

All qualitative examples shown are generated using the model variant that was trained with the mesh-based control signal, using the smaller context-length SVD version, conditioned on the CLIP embedding of the first frame and receiving both the input and the control signal.

Validation ID

Figure 21 and Figure 22 show two generated videos on our ID validation split. As training progresses, we see that the model is able to accurately model unseen dynamics for these objects.

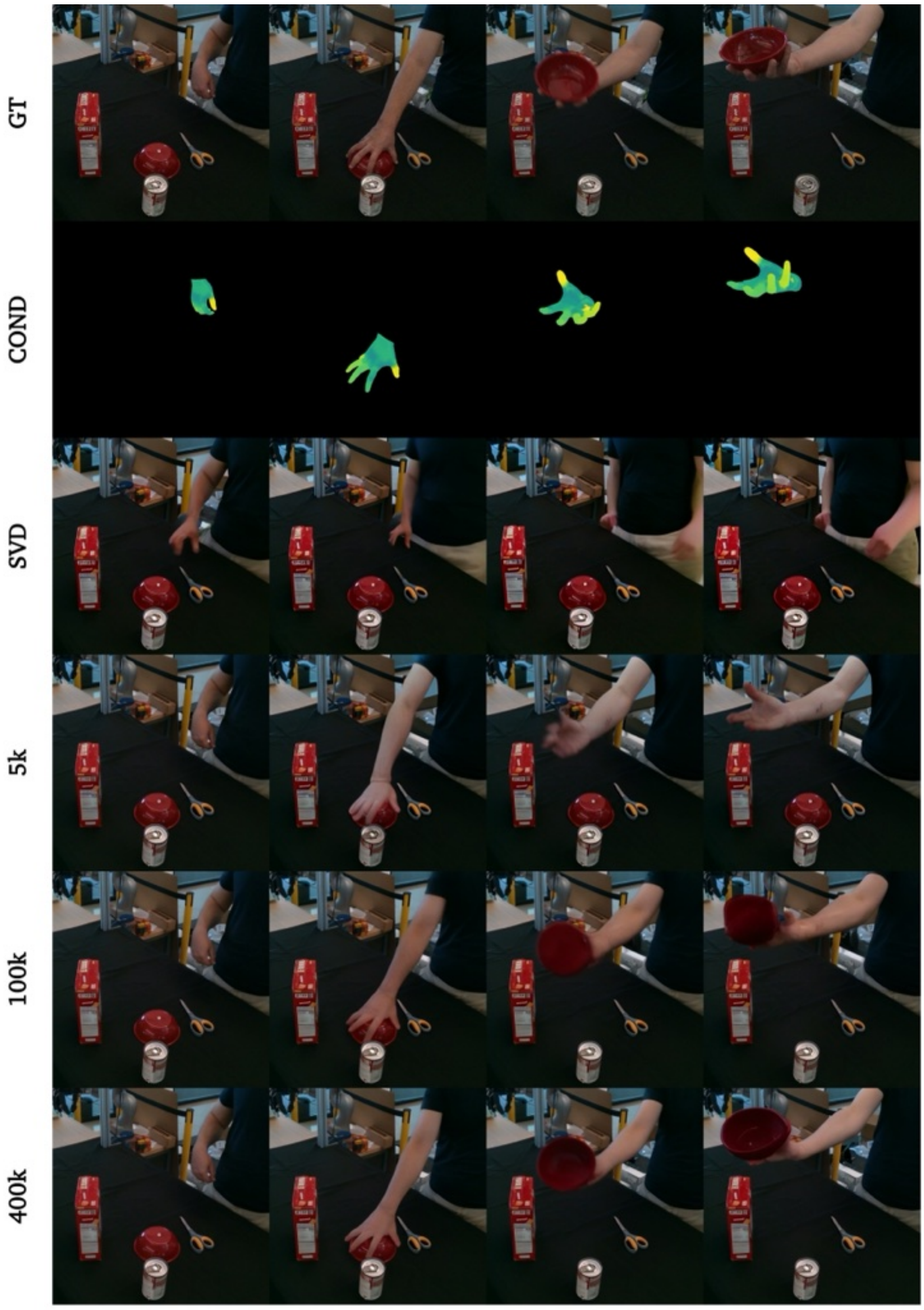


Figure 21: Bowl - Success case. One particularly difficult sequence in which a bowl is turned upside down. GT = Ground truth, COND = Conditioning.

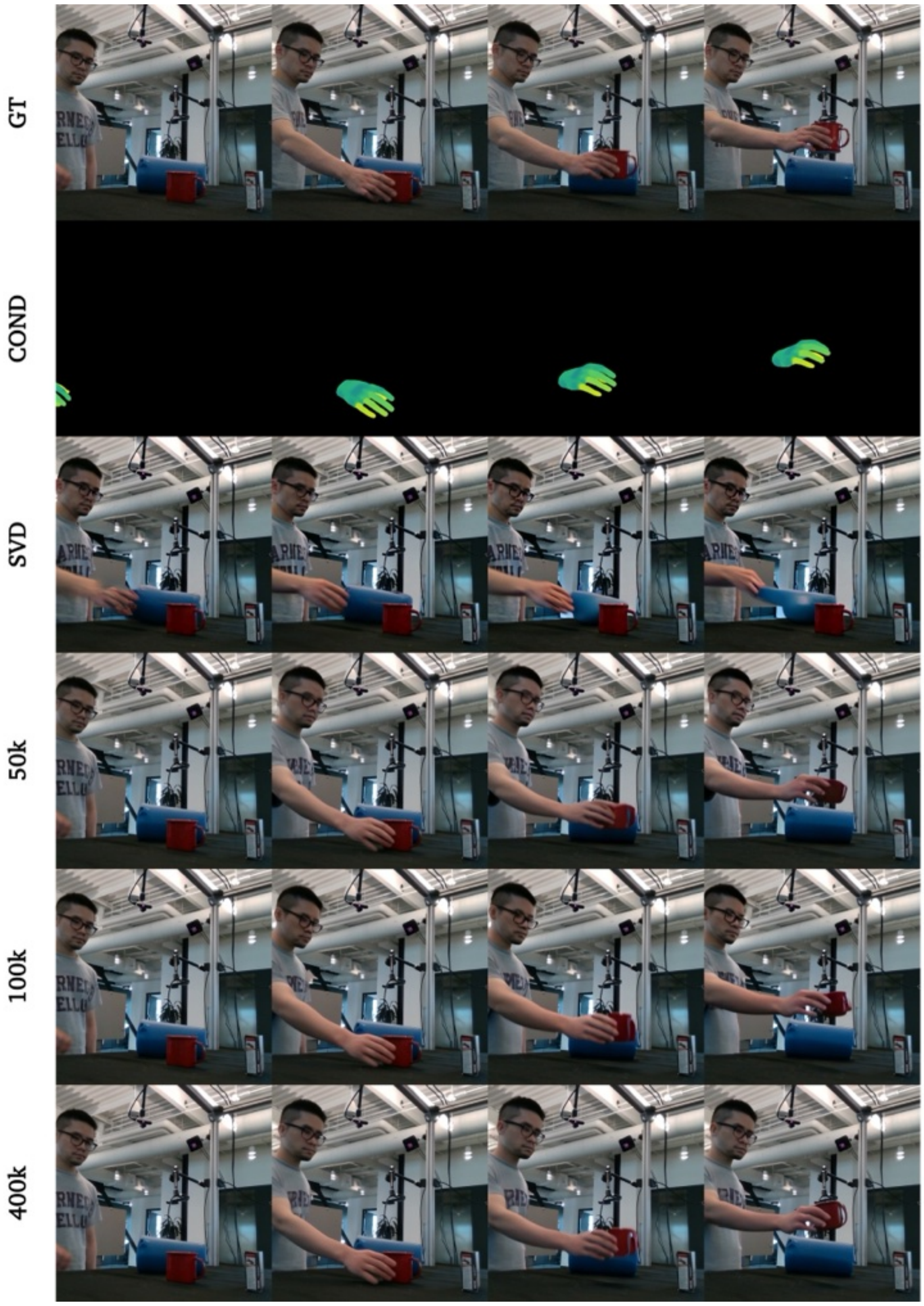


Figure 22: Mug - Success case. One sequence in which a mug is picked up and brought up. GT = Ground truth, COND = Conditioning.

Validation OOD

In Figure 23 and Figure 24, we show two generated videos on our OOD validation split. Over the course of training, we see that the model is able to generalize the control signal to the generation of the human as well as to the object. In general, the overall shape and colour of the object is maintained, however, object-specific details are often lost.

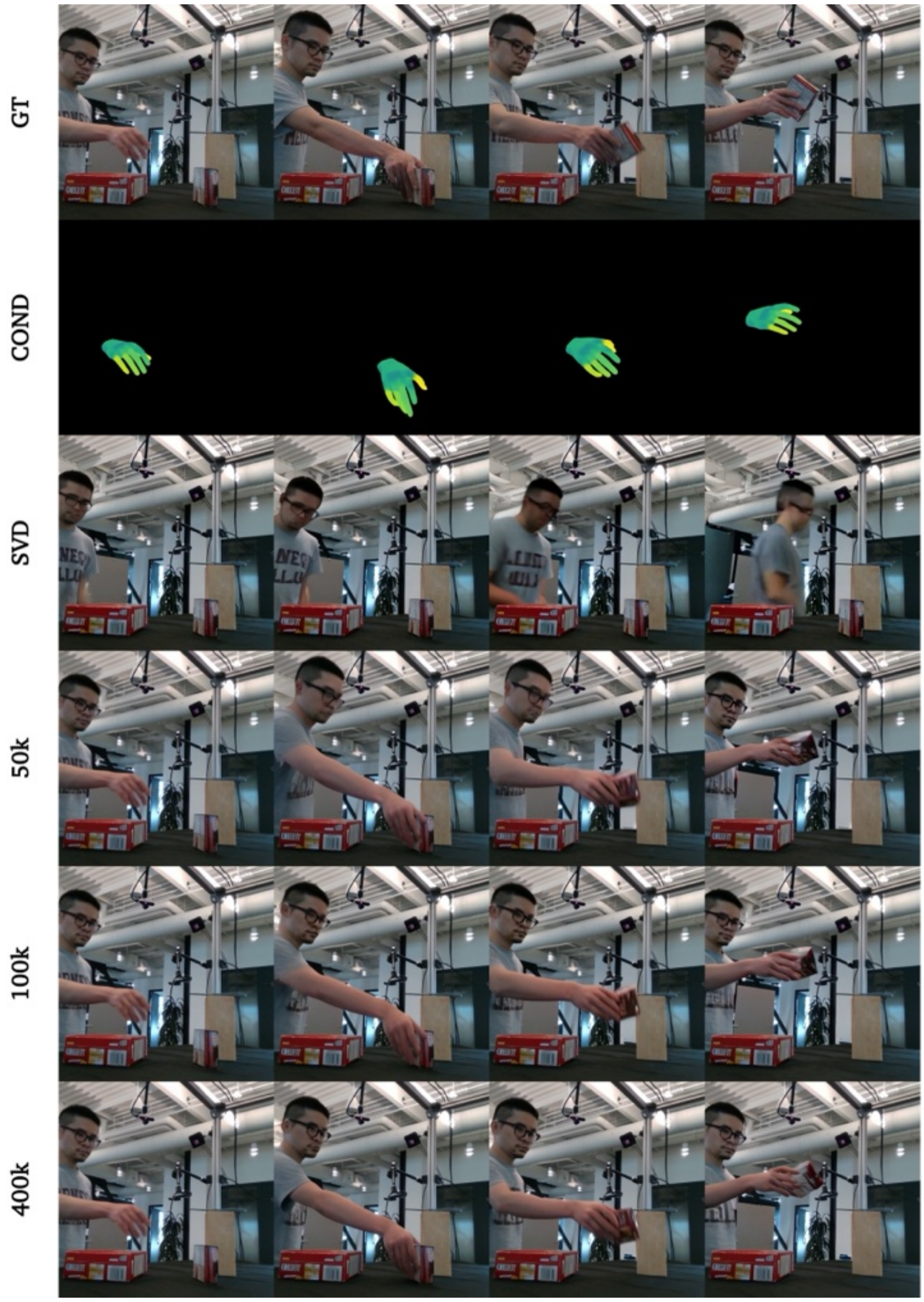


Figure 23: Pudding box - Success case. By the end of training, the model is able to grasp the pudding box and maintain its overall shape throughout the motion. GT = Ground truth, COND = Conditioning.

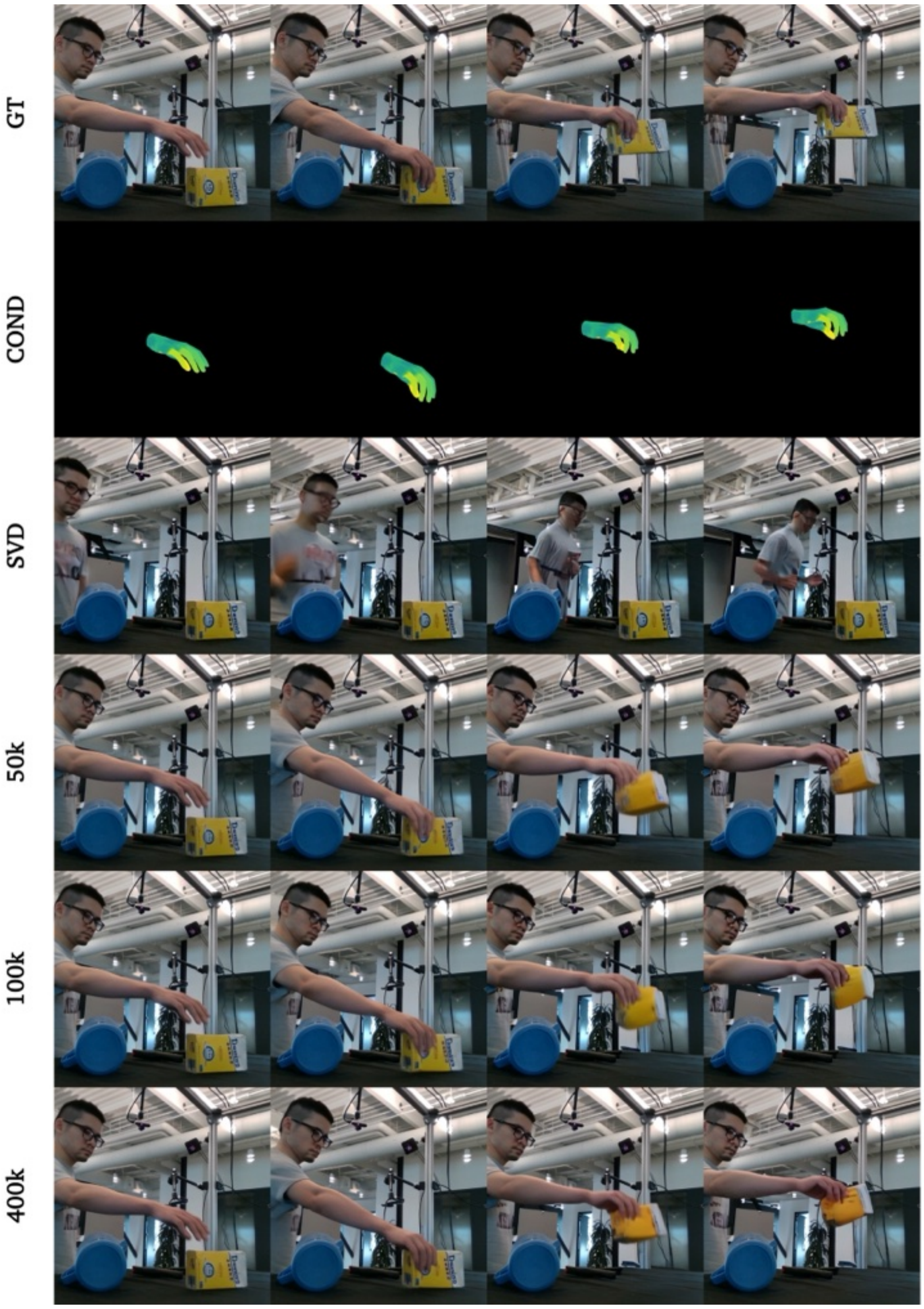


Figure 24: Sugar box - Success case. By the end of training, the model is able to grasp the sugar box and maintain its overall shape throughout the motion. GT = Ground truth, COND = Conditioning.

Failure cases

In Figure 25, Figure 26, and Figure 27 we show three failure cases due to (1) objects that are outside of the frame, (2) the model does not recognize the correct orientation of the hand in the control signal, and (3) the control signal does not always generalize to unseen objects, with the consequence that they are not picked up.

Figure 25: Drill - Failure case. While we removed all sequences that do not have a hand in the first frame, we did not do the same for objects that might be outside of the frame. GT = Ground truth, COND = Conditioning.

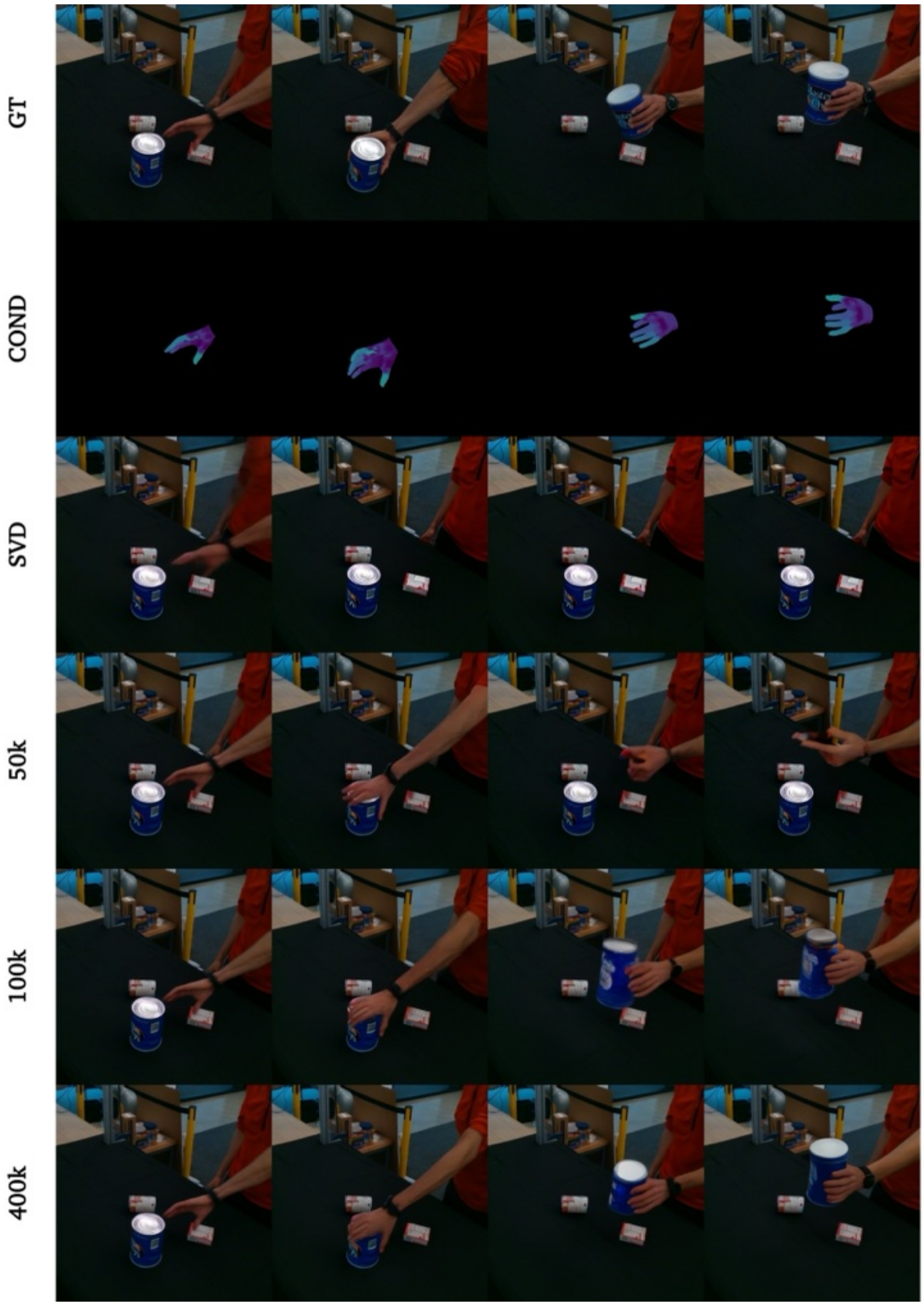


Figure 26: Masterchef - Failure case. While the hand follows the control signal spatially, the model does not recognize the orientation of the hand. GT = Ground truth, COND = Conditioning.

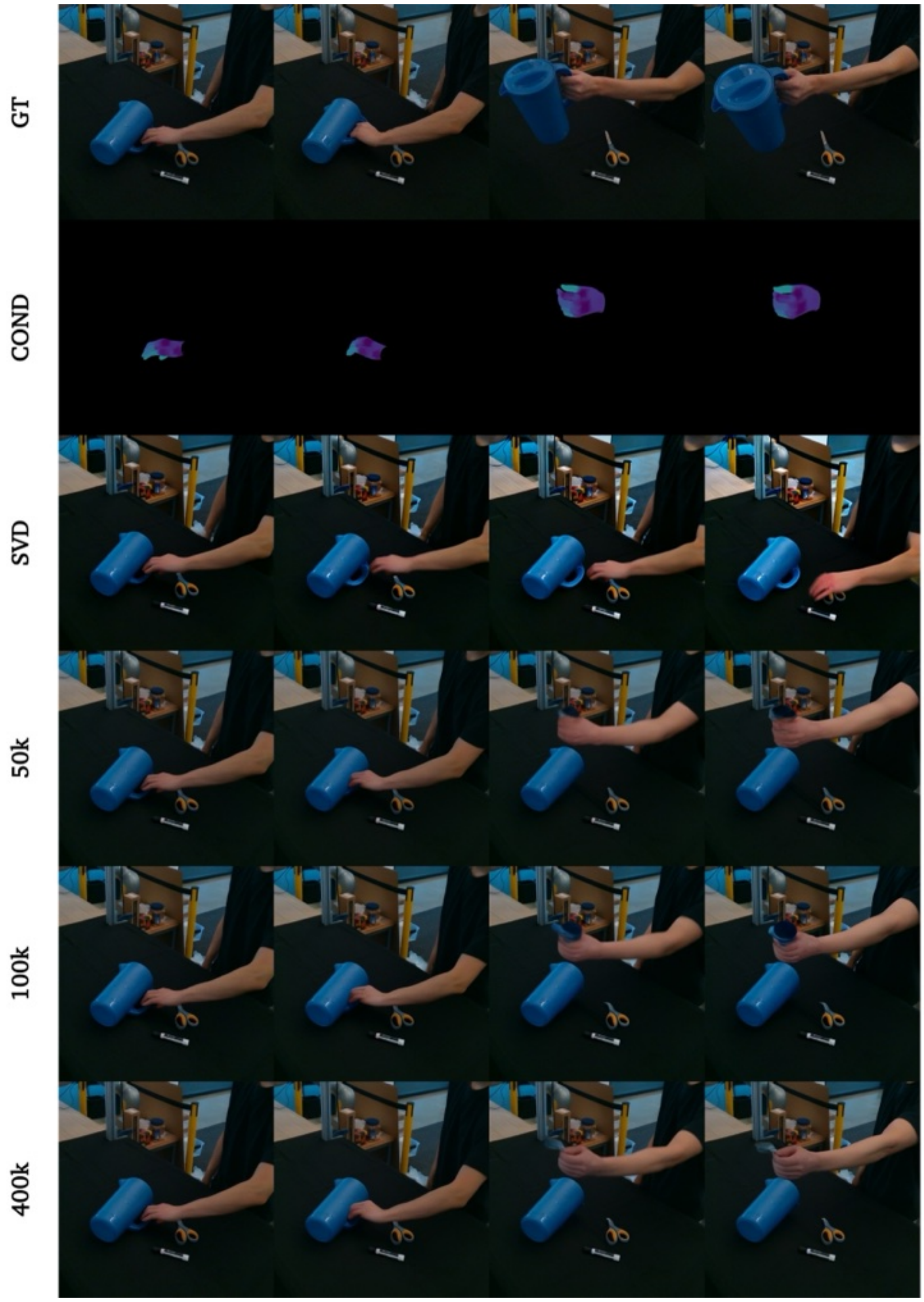


Figure 27: Pitcher - Failure case. Sometimes for objects in the OOD validation split, the object is not picked up. GT = Ground truth, COND = Conditioning.

6 Conclusion

In this master's thesis, we explored the novel task of synthesizing human-object interactions from a single image, guided by a control signal that dictates human motion. The core of our approach lies in leveraging the capabilities of large video foundation models, particularly Stable Video Diffusion (SVD), to act as both a simulator and a neural renderer. This approach bypasses the need for explicit 3D modeling, physics simulations, and rendering pipelines, capitalizing on the rich implicit representations learned by these models from vast video data.

Our experimental results demonstrate the feasibility of this approach, showcasing the model's ability to generate high-fidelity, dynamic human-object interactions. The generated videos exhibit plausible object dynamics in response to the controlled human movements, even without explicit object motion control signals. This underscores the potential of foundation models to capture and reproduce complex real-world interactions.

However, our exploration also highlighted several challenges and areas for future research. The model's performance is sensitive to the design of control signals, information flow, and training strategies. The choice of control signal representation, the level of detail provided to the ControlNet encoder, and the balance between generalization and overfitting all played crucial roles in achieving optimal results.

Future work could delve deeper into these aspects, exploring more diverse and large-scale datasets to enhance the model's generalization capabilities. Additionally, incorporating more advanced 3D control signal representations could offer finer control over the generated interactions. Furthermore, integrating explicit 3D representations with the implicit knowledge of video models could lead to even more realistic and controllable human-object interaction synthesis. The combination of video models with explicit representations could ensure the consistency of objects over time, even in complex interactions. It could also allow for the incorporation of prior knowledge about how humans typically interact with objects, leading to more plausible and intuitive generations. Moreover, text could be used as guidance to control the physical properties of objects in the scene, enabling the generation of interactions that respond realistically to varying object properties.

The evaluation of generated videos, particularly in the context of human-object interaction, presents a significant challenge. While metrics like FVD offer a holistic assessment of video quality, and image quality metrics provide insights into individual frame fidelity, they fall short in capturing the nuances of temporal consistency and the realism of object dynamics. The lack of an appropriate metric that quantifies the distribution of motion and geometrical consistency of objects over time limits our ability to fully assess the model's understanding of object dynamics. The development of such a metric, potentially leveraging accurate point tracking methods, could provide a more comprehensive evaluation of video generation models, particularly for tasks involving complex object dynamics and interactions.

In conclusion, this thesis contributes to the growing field of human-object interaction synthesis by demonstrating the potential of large video foundation models. While challenges remain, the results presented here pave the way for future research that could lead to even more sophisticated and realistic generation of human-object interactions in video. The ability to synthesize such interactions has broad implications for various applications, including animation, virtual reality, and robotics, where creating realistic and engaging human-object interactions is paramount.

References

[1] Video generation models as world simulators. February 2024. OpenAI.

[2] Mohammad Babaeizadeh, Chelsea Finn, Dumitru Erhan, Roy H Campbell, and Sergey Levine. Stochastic variational video prediction. *arXiv preprint arXiv:1710.11252*, 2017.

[3] Bharat Lal Bhatnagar, Xianghui Xie, Ilya A Petrov, Cristian Sminchisescu, Christian Theobalt, and Gerard Pons-Moll. Behave: Dataset and method for tracking human object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15935–15946, 2022.

[4] Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, et al. Stable video diffusion: Scaling latent video diffusion models to large datasets. *arXiv preprint arXiv:2311.15127*, 2023.

[5] Andreas Blattmann, Robin Rombach, Huan Ling, Tim Dockhorn, Seung Wook Kim, Sanja Fidler, and Karsten Kreis. Align your latents: High-resolution video synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22563–22575, 2023.

[6] Federica Bogo, Angjoo Kanazawa, Christoph Lassner, Peter Gehler, Javier Romero, and Michael J Black. Keep it smpl: Automatic estimation of 3d human pose and shape from a single image. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11-14, 2016, Proceedings, Part V 14*, pages 561–578. Springer, 2016.

[7] Andrew Brock, Jeff Donahue, and Karen Simonyan. Large scale gan training for high fidelity natural image synthesis. *arXiv preprint arXiv:1809.11096*, 2018.

[8] Tim Brooks, Janne Hellsten, Miika Aittala, Ting-Chun Wang, Timo Aila, Jaakko Lehtinen, Ming-Yu Liu, Alexei Efros, and Tero Karras. Generating long videos of dynamic scenes. *Advances in Neural Information Processing Systems*, 35:31769–31781, 2022.

[9] Joao Carreira and Andrew Zisserman. Quo vadis, action recognition? a new model and the kinetics dataset. In *proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 6299–6308, 2017.

[10] Lluis Castrejon, Nicolas Ballas, and Aaron Courville. Improved conditional vrnns for video prediction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 7608–7617, 2019.

[11] Yu-Wei Chao, Wei Yang, Yu Xiang, Pavlo Molchanov, Ankur Handa, Jonathan Tremblay, Yashraj S Narang, Karl Van Wyk, Umar Iqbal, Stan Birchfield, et al. Dexycb: A benchmark for capturing hand grasping of objects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9044–9053, 2021.

[12] Ching-Hang Chen and Deva Ramanan. 3d human pose estimation= 2d pose estimation+ matching. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 7035–7043, 2017.

[13] Haoxin Chen, Menghan Xia, Yingqing He, Yong Zhang, Xiaodong Cun, Shaoshu Yang, Jinbo Xing, Yaofang Liu, Qifeng Chen, Xintao Wang, et al. Videocrafter1: Open diffusion models for high-quality video generation. *arXiv preprint arXiv:2310.19512*, 2023.

[14] Tianqi Chen, Bing Xu, Chiyuan Zhang, and Carlos Guestrin. Training deep nets with sublinear memory cost. *arXiv preprint arXiv:1604.06174*, 2016.

[15] Joseph Cho, Fachrina Dewi Puspitasari, Sheng Zheng, Jingyao Zheng, Lik-Hang Lee, Tae-Ho Kim, Choong Seon Hong, and Chaoning Zhang. Sora as an agi world model? a complete survey on text-to-video generation. *arXiv preprint arXiv:2403.05131*, 2024.

[16] Aidan Clark, Jeff Donahue, and Karen Simonyan. Adversarial video generation on complex datasets. *arXiv preprint arXiv:1907.06571*, 2019.

[17] Emily Denton and Rob Fergus. Stochastic video generation with a learned prior. In *International conference on machine learning*, pages 1174–1183. PMLR, 2018.

[18] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021.

[19] Bin Duan, Wei Wang, Hao Tang, Hugo Latapie, and Yan Yan. Cascade attention guided residue learning gan for cross-modal translation. In *2020 25th International Conference on Pattern Recognition (ICPR)*, pages 1336–1343. IEEE, 2021.

[20] Kemal Erdem. Step by step visual introduction to diffusion models. *Personal Website*, 11 2023. Accessed: date-of-access.

[21] Patrick Esser, Robin Rombach, and Bjorn Ommer. Taming transformers for high-resolution image synthesis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12873–12883, 2021.

[22] Zicong Fan, Omid Taheri, Dimitrios Tzionas, Muhammed Kocabas, Manuel Kaufmann, Michael J Black, and Otmar Hilliges. Arctic: A dataset for dexterous bimanual hand-object manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12943–12954, 2023.

[23] Haiwen Feng, Zheng Ding, Zhihao Xia, Simon Niklaus, Victoria Abrevaya, Michael J Black, and Xuaner Zhang. Explorative inbetweening of time and space. *arXiv preprint arXiv:2403.14611*, 2024.

[24] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial networks. *Communications of the ACM (CACM)*, 63(11):139–144, 2020.

[25] Shreyas Hampali, Mahdi Rad, Markus Oberweger, and Vincent Lepetit. Honnotate: A method for 3d annotation of hand and object poses. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3196–3206, 2020.

[26] Yana Hasson, Gul Varol, Dimitrios Tzionas, Igor Kalevatykh, Michael J Black, Ivan Laptev, and Cordelia Schmid. Learning joint reconstruction of hands and manipulated objects. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11807–11816, 2019.

[27] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.

[28] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Conference on Neural Information Processing Systems (NeurIPS)*, 33:6840–6851, 2020.

[29] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

[30] Hezhen Hu, Weilun Wang, Wengang Zhou, and Houqiang Li. Hand-object interaction image generation. *Advances in Neural Information Processing Systems*, 35:23805–23817, 2022.

[31] Yinlin Hu, Joachim Hugonot, Pascal Fua, and Mathieu Salzmann. Segmentation-driven 6d object pose estimation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3385–3394, 2019.

[32] Yinghao Huang, Omid Taheri, Michael J Black, and Dimitrios Tzionas. Intercap: Joint markerless 3d tracking of humans and objects in interaction. In *DAGM German Conference on Pattern Recognition*, pages 281–299. Springer, 2022.

[33] Hao Jiang. 3d human pose reconstruction using millions of exemplars. In *2010 20th International Conference on Pattern Recognition*, pages 1674–1677. IEEE, 2010.

[34] Angjoo Kanazawa, Michael J Black, David W Jacobs, and Jitendra Malik. End-to-end recovery of human shape and pose. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 7122–7131, 2018.

[35] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. *Advances in neural information processing systems*, 35:26565–26577, 2022.

[36] Tero Karras, Samuli Laine, and Timo Aila. A style-based generator architecture for generative adversarial networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4401–4410, 2019.

[37] Diederik P Kingma and Max Welling. Auto-encoding variational bayes. *International Conference on Learning Representations (ICLR)*, 2013.

[38] Yitong Li, Martin Min, Dinghan Shen, David Carlson, and Lawrence Carin. Video generation from text. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.

[39] Pauline Luc, Aidan Clark, Sander Dieleman, Diego de Las Casas, Yotam Doron, Albin Cassirer, and Karen Simonyan. Transformation-based adversarial video prediction on large-scale data. *arXiv preprint arXiv:2003.04035*, 2020.

[40] Wan-Duo Kurt Ma, John P Lewis, and W Bastiaan Kleijn. Trailblazer: Trajectory control for diffusion-based video generation. *arXiv preprint arXiv:2401.00896*, 2023.

[41] Julieta Martinez, Rayat Hossain, Javier Romero, and James J Little. A simple yet effective baseline for 3d human pose estimation. In *Proceedings of the IEEE international conference on computer vision*, pages 2640–2649, 2017.

[42] Paulius Micikevicius, Sharan Narang, Jonah Alben, Gregory Diamos, Erich Elsen, David Garcia, Boris Ginsburg, Michael Houston, Oleksii Kuchaiev, Ganesh Venkatesh, et al. Mixed precision training. *arXiv preprint arXiv:1710.03740*, 2017.

[43] Raphaël Millière. Are video generation models world simulators?, 2024.

[44] Gaurav Mittal, Tanya Marwah, and Vineeth N Balasubramanian. Sync-draw: Automatic video generation using deep recurrent attentive architectures. In *Proceedings of the 25th ACM international conference on Multimedia*, pages 1096–1104, 2017.

[45] Andres Munoz, Mohammadreza Zolfaghari, Max Argus, and Thomas Brox. Temporal shift gan for large scale video generation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 3179–3188, 2021.

[46] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *International conference on machine learning*, pages 8162–8171. PMLR, 2021.

[47] Yingwei Pan, Zhaofan Qiu, Ting Yao, Houqiang Li, and Tao Mei. To create what you tell: Generating videos from captions. In *Proceedings of the 25th ACM international conference on Multimedia*, pages 1789–1798, 2017.

[48] Kiru Park, Timothy Patten, and Markus Vincze. Pix2pose: Pixel-wise coordinate regression of objects for 6d pose estimation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 7668–7677, 2019.

[49] Georgios Pavlakos, Xiaowei Zhou, Konstantinos G Derpanis, and Kostas Daniilidis. Coarse-to-fine volumetric prediction for single-image 3d human pose. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 7025–7034, 2017.

[50] Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville. Film: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.

[51] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021.

[52] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. *Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695, 2022.

[53] Javier Romero, Dimitrios Tzionas, and Michael J Black. Embodied hands: Modeling and capturing hands and bodies together. *arXiv preprint arXiv:2201.02610*, 2022.

[54] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18*, pages 234–241. Springer, 2015.

[55] Masaki Saito, Eiichi Matsumoto, and Shunta Saito. Temporal generative adversarial nets with singular value clipping. In *Proceedings of the IEEE international conference on computer vision*, pages 2830–2839, 2017.

[56] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *International conference on machine learning*, pages 2256–2265. PMLR, 2015.

[57] Ximeng Sun, Huijuan Xu, and Kate Saenko. A two-stream variational adversarial network for video generation. *arXiv preprint arXiv:1812.01037*, 1:12, 2018.

[58] Ximeng Sun, Huijuan Xu, and Kate Saenko. Twostreamvan: Improving motion modeling in video generation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 2744–2753, 2020.

[59] Bugra Tekin, Pablo Márquez-Neila, Mathieu Salzmann, and Pascal Fua. Learning to fuse 2d and 3d image cues for monocular body pose estimation. In *Proceedings of the IEEE international conference on computer vision*, pages 3941–3950, 2017.

[60] Bugra Tekin, Sudipta N Sinha, and Pascal Fua. Real-time seamless single shot 6d object pose prediction. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 292–301, 2018.

[61] Yu Tian, Jian Ren, Menglei Chai, Kyle Olszewski, Xi Peng, Dimitris N Metaxas, and Sergey Tulyakov. A good image generator is what you need for high-resolution video synthesis. *arXiv preprint arXiv:2104.15069*, 2021.

[62] Sergey Tulyakov, Ming-Yu Liu, Xiaodong Yang, and Jan Kautz. Mocogan: Decomposing motion and content for video generation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1526–1535, 2018.

[63] Thomas Unterthiner, Sjoerd Van Steenkiste, Karol Kurach, Raphael Marinier, Marcin Michalski, and Sylvain Gelly. Towards accurate generative models of video: A new metric & challenges. *arXiv preprint arXiv:1812.01717*, 2018.

[64] Aaron Van Den Oord, Oriol Vinyals, et al. Neural discrete representation learning. *Advances in neural information processing systems*, 30, 2017.

[65] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Conference on Neural Information Processing Systems (NeurIPS)*, 30, 2017.

[66] Ruben Villegas, Jimei Yang, Seunghoon Hong, Xunyu Lin, and Honglak Lee. Decomposing motion and content for natural video sequence prediction. *arXiv preprint arXiv:1706.08033*, 2017.

[67] Vikram Voleti, Chun-Han Yao, Mark Boss, Adam Letts, David Pankratz, Dmitry Tochilkin, Christian Laforte, Robin Rombach, and Varun Jampani. Sv3d: Novel multi-view synthesis and 3d generation from a single image using latent video diffusion. *arXiv preprint arXiv:2403.12008*, 2024.

[68] Carl Vondrick, Hamed Pirsiavash, and Antonio Torralba. Generating videos with scene dynamics. *Advances in neural information processing systems*, 29, 2016.

[69] He Wang, Srinath Sridhar, Jingwei Huang, Julien Valentin, Shuran Song, and Leonidas J Guibas. Normalized object coordinate space for category-level 6d object pose and size estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2642–2651, 2019.

[70] Jiawei Wang, Yuchen Zhang, Jiaxin Zou, Yan Zeng, Guoqiang Wei, Liping Yuan, and Hang Li. Boximator: Generating rich and controllable motions for video synthesis. *arXiv preprint arXiv:2402.01566*, 2024.

[71] Yaohui Wang, Piotr Bilinski, Francois Bremond, and Antitza Dantcheva. G3an: Disentangling appearance and motion for video generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5264–5273, 2020.

[72] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing*, 13(4):600–612, 2004.

[73] Zhouxia Wang, Ziyang Yuan, Xintao Wang, Yaowei Li, Tianshui Chen, Menghan Xia, Ping Luo, and Ying Shan. Motionctrl: A unified and flexible motion controller for video generation. In *ACM SIGGRAPH 2024 Conference Papers*, pages 1–11, 2024.

[74] Yu Xiang, Tanner Schmidt, Venkatraman Narayanan, and Dieter Fox. Posecnn: A convolutional neural network for 6d object pose estimation in cluttered scenes. *arXiv preprint arXiv:1711.00199*, 2017.

[75] Wei Xie, Zhipeng Yu, Zimeng Zhao, Binghui Zuo, and Yangang Wang. Hmdo: Markerless multi-view hand manipulation capture with deformable objects. *Graphical Models*, 127:101178, 2023.

[76] Wilson Yan, Yunzhi Zhang, Pieter Abbeel, and Aravind Srinivas. Videogpt: Video generation using vq-vae and transformers. *arXiv preprint arXiv:2104.10157*, 2021.

[77] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3836–3847, 2023.

[78] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 586–595, 2018.

A Training hyperparameters

Category	Hyperparameter	Setting
Control	ControlNet weight initialization	U-Net
	Conditioning dropout probability	0.1
Data	Num frames	14
	Channels	3
	Height	512
	Width	512
Diffusion [4, 35]	$\log \sigma$	$\mathcal{N}(0.7, 1.6)$
	σ_{data}	1
	$\log \sigma$	$\mathcal{N}(P_{\text{mean}}, P_{\text{std}}^2)$
	$c_{\text{skip}}(\sigma)$	$(\sigma^2+1)^{-1}$
	$c_{\text{out}}(\sigma)$	$\frac{-\sigma}{\sqrt{\sigma^2+1}}$
	$c_{\text{in}}(\sigma)$	$\frac{1}{\sqrt{\sigma^2+1}}$
	$c_{\text{noise}}(\sigma)$	$0.25 \log \sigma$
	Noise augmentation strength	$\mathcal{N}(-3, 0.5)$
Batch and memory	Number of GPUs	4
	Per GPU batch size	2
	Gradient accumulation steps	2
	Effective batch size	16
	GPU type	NVIDIA H100 80GB GPU
	Gradient checkpointing	True
	Mixed precision	fp16
Training	Training iterations	$400k$
	Learning rate	$3e-5$
	Adam beta1	0.9
	Adam beta2	0.999
	Adam weight decay	$1e-2$
	Adam epsilon	$1e-08$
	Learning rate scheduler	Constant
	Learning rate warmup steps	500
	Scale learning rate	False
	Max grad norm	1.0
	Checkpointing steps	1000
	Seed	1

Table 9: Training hyperparameters. For training ControlNet

B CLIP Object Descriptions

Object	Textual description
Banana	an image of a yellow banana
Bleach cleanser	an image of a white bottle of bleach cleanser
Bowl	an image of a brown bowl
Cracker box	an image of a red cracker box
Extra large clamp	an image of a large black clamp
Foam brick	an image of a red foam brick
Gelatin box	an image of a red gelatin box
Large marker	an image of a white and black marker
Master chef can	an image of a blue master chef coffee can
Mug	an image of a red mug
Mustard bottle	an image of a yellow mustard bottle
Potted meat can	an image of a metal potted meat can
Scissors	an image of a black and orange pair of scissors
Tomato soup can	an image of a white and red tomato soup can
Tuna fish can	an image of a blue metal tuna fish can
Wood block	an image of a light coloured block of wood
Pitcher base	an image of a blue plastic water pitcher base
Power drill	an image of an orange and black power drill
Pudding box	an image of a brown pudding box
Sugar box	an image of a yellow sugar box

Table 10: Object descriptions. Textual descriptions employed for the CLIP embedding of each object.

7 Review

7.1 Examiner

Question	Examiner
Summary	This thesis investigates the problem of video generation, specifically generating frames given a starting frame and control information. Foundation models in the video domain are explored to test whether they are capable of performing such conditional generation. Experiments show that different types of controls can be incorporated to guide the generation process, making for an interesting direction in the video generation landscape.
Strengths	**[S1]** The idea to generate videos with foundation models and control signals is interesting yet logical. Fully unconditioned generation is likely to misrepresent physics, rigid motions, and human anatomy. By having strict priors such as skeletons, meshes, and semantics, generations are more likely to be realistic. **[S2]** The proposed task is new and the accompanying method architecture combines video diffusion with controlnet to align the video signals with the priors. Overall a complex and well-motivated method. **[S3]** The experiments show both quantitative and qualitative analyses. Especially the qualitative analyses help with understanding the potential of different controls.
Weaknesses	**[W1]** The writing of the thesis could have been improved. This is not about the wording itself, but mostly about the lack of compression. The thesis over the page limit and this is a direct result of the overload of information. For example, the thesis boasts 14 (!) pages of related work and background. This is not a strength, but a failure to condense information. Instead of distilling the essentials to the reader, the reader is just given all information. Similarly, the introduction and conclusions are far lengthier than needed. It would have been better to get to the point, in its current format it is difficult for the reader what is exactly new and different here given all the superfluous information. **[W2]** Looking at the metrics in Tables 5 to 8, I do not see higher perceptual scores when adding the controls compared to SVD only. Is this correct and as expected? Does this mean that controls do not lead to better perceptual quality of generated videos?

Weaknesses (cont.)	[W3] How does the proposed approach related to existing approaches? The abstract and introduction state limitations of existing directions, but how well does the proposed approach score in terms of perceptual metrics or qualitatively? In other words: what does this paper add empirically compared to the many existing papers? [W4] The analysis in 5.5 is very interesting, but I'm not sure I agree with its conclusions. Section 5.5 concludes that low-frequency and homogeneous objects tend to do better. But wood blocks are high-frequency while mugs are often small and not homogeneous. Yet both work well. Could it be that the proposed approach favors rigid objects? E.g., a banana is subject to change when pealing it. A mug is small but rigid, making it more predictable for generation. I am curious about the opinion of the student.
Initial indication	Very Good
Address in rebuttal	Overall, this is a thesis with a new task, complex method, and cool outcomes. the writing and organization of thesis could have been better, while there are multiple open questions in the experiments and comparisons. Therefore my initial rating is in between very good and excellent. I would like to know the opinion of the student regarding the raised points 2.1-2.3 above.

7.2 Other reader

Question	Examiner
Summary	The thesis focuses on the novel task of video synthesis of human-object interaction, with a specific focus on generating object dynamics while conditioning generation only on human motion. This is a new task, so this work is an early exploration, and has good prospects for impact after extensions. The thesis presents a method that extends (finetunes) Stable Video Diffusion with controllability through a ControlNet model that is conditioned on human motion signals (and potentially on object 'identity'). The DexYCB dataset is used for training and evaluation. Various design choices are evaluated in an objective benchmark. The discussion is transparent and educational. The results look really encouraging, and I encourage the author to extend this work further towards a submission at a top-tier conference. The sections below discuss 5 strengths and 5 suggestions for improvement.

Strengths	[S1] The task is new, and is related to the general research topic of generative-AI for videos, which is hot. This means that, for successful methods, prospects of impact are high. [S2] The method details and experimental benchmark are clear enough to ensure reproducibility (although writing could benefit by a polishing pass). Figures are insightful, and the tables (and appendix) that summarize technical steps help reproducibility (ideally code should be released as well – see question for the rebuttal at the bottom). [S3] Many qualitative results are provided. This gives confidence to the reader, by providing deep insights into the results. [S4] I appreciate the transparent presentation of both 'success cases' and 'failure cases' and the discussion on hypothesized reasons for the latter. This informs the reader well about future-work prospects. [S5] Although writing could improve (to be less verbose/repetitive), the overall structure is well designed, so the reader can easily navigate the manuscript and find desired information. I also clarified several technical things in my head, which were a bit fuzzy or too high-level earlier for me, so I found the manuscript educative (but again, there is room for improvement).
Weaknesses	[W1] The manuscript reads well, but would benefit from a polishing pass. That is, some sections read well (especially the technical ones), while other sections are a bit verbose (introductory ones) or a bit repetitive (results). Sometimes implementation details also get 'prioretized' in the text over more 'basic' information for the reader. This should not be considered a deal breaker, but is an indication that writing needs extra quality time ;) On another note, though, there is a good attention to detail with clean and consistent bibliographic information, as well as tables of contents, which is a good sign (so, things are 'just' a matter of devoting extra time). The command of the English language is also good. [W2] I appreciate the discussion that quantitative metrics are imperfect for image/video synthesis (consequently, and naturally, sometimes no strong conclusion can be drawn). Indeed, future work on this would help the community tremendously. However, for this exact reason, the thesis would benefit tremendously by conducting a perceptual study, in which human participants rate the realism of generated videos in comparison to real ones. [W3] The text motivates the use of GT signals for conditioning [page 19, bottom paragraph, 'the human and the control signal are always image aligned']. However, it would still be nice to see (even an early version of) a sensitivity analysis. For the rebuttal, empirically, how robust is the model to (various levels of) noise in the conditioning signal? Any hints for future work?

Weaknesses (cont.)	**[W4]** The current steps employ (only) the DexYCB dataset. Although this is fine, for next steps it would help to employ an additional dataset – this would make the author's point stronger. **[W5]** Some typos or awkward phrases/presentation (as natural in any text):

Where	What
Page 10, Fig. 6, top bubble	'denoising' –> 'adding noise to'
Page 14	Double word: 'Figure Figure 8'
Page 16	Reads awkward: 'hint at this severely'
Page 16	Why not reference Fig. 1? 'in the teaser examples'
Page 16	'overtime' –> 'over time'
Page 25	Reads awkward 'standard 3D pose model'
Page 25, Fig. 13	Annotation is not 'crowd sourced' because page 26 explains that this is done through fitting, i.e. 'minimizing a loss function [...]'
Page 26	'wide ranges of motion of the hand' –> Reads awkward
Fig. 12	When the figure is first presented, the 'object description' box (top-right corner) comes a bit as a surprise to the reader, because it has not been motivated earlier.
Page 20, 29, 32	The skeleton conditioning is discussed as 'dense' information in page 20, but as 'sparse' information in page 29 and 32.

Initial indication	Very Good
Address in rebuttal	See W3 above. Are there any plans to release the model and code for public use (e.g. for non-commercial research)? Moreover, are there any plans to submit this work to a conference? If yes, then, what would roughly be the main extensions over the current work?

8 Rebuttal

8.1 Examiner

I thank the examiner for their time, constructive comments and appreciations of the strengths of this work such as 'interesting direction in the video generation landscape', 'complex and well motivated method' and general comments on the usefulness of the qualitative analysis. The intention behind my thesis was to create a comprehensive and easily reproducible guide for newcomers in the field, with a large focus on training specifications and motivations. In striving for thoroughness, I acknowledge that the amount of information may have obscured the core innovations to some extent. Moving forward, I will aim to strike a better balance between detail and conciseness by condensing information better and moving more irrelevant details to the appendix. I value the feedback on this aspect and hope that the overall intent of ensuring completeness and reproducibility is appreciated. Hereunder, I will answer your questions.

[W1] Perceptual metrics: The examiner is correct that Tables 5-8 in the thesis do not show a considerable difference in perceptual scores. This is because they do not show a direct comparison between the proposed SVD+ControlNet approach and the vanilla (uncontrolled) SVD, but between two separately trained SVD+ControlNet versions. For instance, in Table 5, I report results on two versions of SVD+ControlNet, one that initializes the weights with the 14-frame context length SVD model and the other one with the 25-frame context length. All Tables 5-8 follow this template and I actually never report the metrics for vanilla SVD out-of-the-box.

Model	FVD ↓	MSE ↓	LPIPS ↓	PSNR ↑	SSIM ↑
Vanilla SVD	370.864	0.217	0.294	18.087	0.689

Table 13: Performance of Vanilla SVD on the out-of-distribution validation split

To address the examiner's question on a comparison between Vanilla SVD and the proposed method, I ran Vanilla SVD on the "Val OOD" split. The results are presented in Table 13. When compared to the metrics in 5-8 in the thesis, it can be seen that the proposed method substantially improves all metrics across the board. First of all, the proposed method substantially outperforms Vanilla SVD in perceptual scores. Secondly, the high FVD score for vanilla SVD highlights the difficulty of generating realistic and plausible human-object interactions that are distributed closely to the ground truth, without any guidance or control. Lastly, the significantly higher object MSE is expected, since Vanilla SVD has no control signal that might indicate where the object should move. The additional experiment presented clearly demonstrates the superior performance of the proposed approach in terms of both perceptual quality and interaction control.

[W2] Comparison with existing approaches: The examiner raises a crucial point about the empirical comparison of the proposed video-based HOI approach to the plentiful 3D-based HOI approaches. DexYCB only provides the ground truth appearance of the objects, but not for the humans or the scene. Therefore, many 3D-based HOI approaches on DexYCB focus solely on predicting the 3D pose of the human and object overtime and do not focus on appearance and photo-realistic rendering of the human, object and scene. This makes a comparison of video-based HOI and 3D-based HOI approaches difficult in terms of perceptual scores. For future work, I would choose a dataset that has the ground truth 3D appearance of the human, object and scene available, so that the proposed method would be directly comparable.

I did find one work from Hu et al. [30] that aims to generate HOI images from a rendering of a hand and object pose on DexYCB. While comparable, their method heavily relies on in-painting, access to the ground truth object shapes, and zooms in on the hand and object region, which does make their work not entirely comparable to this work. In May, I contacted the authors for their dataset processing code and/or trained models to run a comparison, however, received no response. For context, their training code is available but requires substantial computational resources, moreover their code for processing the dataset seems incomplete and lacks instructions.

[W3] Analysis of object characteristics: I appreciate the examiner's careful attention to my analysis in Section 5.5 and their insightful proposal that the method may favor rigid objects due to their predictability offers a compelling alternative explanation. For instance, it has been found that due to uncertainty in object trajectory, VAEs for video generation might average multiple possible outcomes resulting in noisy or blurry images. While a valid suggestion, I would like to stress two important factors that I believe support my initial analysis:

1. **Object distance:** Specifically in starting frames, objects in DexYCB are often far from the camera. This distance makes some high-frequency textures, like those on the wooden block, appear more homogeneous. This contrasts with objects like the cracker box or bleach cleanser, which retain more visual complexity at a distance.

2. **Motion blur:** As noted in Section 4.4, DexYCB comes with considerable motion blur. In combination with a MSE loss during training, this can make it challenging for the model to learn about, and to maintain, high-frequency details.

Considering these factors, I believe the method's stronger performance on objects with less complex textures remains a more likely explanation. However, I fully acknowledge the validity of the examiner's suggestion regarding the predictability of objects. Future work could include an investigation whether video diffusion models suffer the same averaging of multiple possible outcomes as VAEs, as well as expanding the range of objects allowing a more comprehensive analysis.

8.2 Other reader

I thank the other reader for their time, constructive comments and appreciations of the strengths of this work such as 'the discussion is transparent and educational', 'the overall structure is well designed', 'figures are insightful' and general comments on the proposition of the study and its reproducibility. I appreciate the suggestion of the other reader to submit to a conference and I would like to mention that I am currently in the works of extending this work and preparing it for a submission. Similar to the examiner, the other reader is correct that the writing of the thesis could have been improved. I underestimated the time it takes to reiterate on the manuscript once it is fully written and in the future will plan out more time for this. Hereunder, I will answer your questions.

[W2] Perceptual study: I appreciate the others reader's suggestion to perform a perceptual study to overcome the current gap in quantitative metrics for image and video synthesis; this would have been a great addition to my thesis. I will note it down for the paper submission. For context, in the paper submission we are also working on leveraging point tracking methods for an additional quantification method of motion in the generated results. The idea would be to sample N number of points on the object, track these over time for both the ground truth and the generated video and use these trajectories to quantify some aspects of the generation and motion. We intend to use it for: (1) comparing the distribution of object motions in ground truth videos and generated videos (2) evaluating whether the generation of the object is consistent enough for points to be tracked.

[W3] Sensitivity analysis: I share the other reader's interest in a sensitivity analysis of the control signal, but I was unfortunately not able to perform an experiment on this within time. Empirically, we did not notice any noise in the control signal when working with DexYCB, however, before settling on this dataset I actually first experimented with the Arctic dataset [22] and the Behave dataset [3] with fully body control signal.

The control signal of Behave is actually quite noisy and we empirically found that noise in the control signal does not severely influence the generated results. For instance, we viewed one sequence in which the foot of the human body model was in an entirely unnatural position in a couple of frames, however, this was ignored by our method and it generated a plausible human foot and sequence overall. I hypothesize that this might be because of (1) the diffusion process is able to correct the mistake at early timesteps by attending to the frames around the frame with the noisy control signal that did receive a clean control signal, (2) the small control signal encoder or the encoder of ControlNet is able to ignore a noisy control signal and passes a 'cleaned' version towards the frozen SVD model. I might even go so far as to say that a noisy control signal might help the method to generalize better and become resilient to more extreme noise. Resilience to training on a noisy control signal would allow me to move away from highly curated datasets with accurate control signals and move towards gathering a large diverse dataset of videos from the internet and track the human movement in these. Such tracking methods generally produce noisy results.

Based on your suggestion, I will plan a simple experiment in which I use an architecture trained on a clean control signal and add noise to the control signal at inference time to quantify its impact on the generated results. Furthermore, I might also train an architecture with a noisy control signal and quantify the impact of a clean and noisy control signal at inference time.

[W4] Additional dataset: I entirely agree with the other reader's suggestion to use an additional dataset, which would hopefully lead to slightly more insightful results, a more robust model and stronger conclusions. Most importantly I am interested in a dataset that has two characteristics: (1) ControlNet does not require a lot of data to finetune, however, it does need to be diverse for the control to generalize. Currently, we trained with a small and skewed dataset, however, for a future dataset I would like to use a small dataset that better covers the full spread of humans, objects, motions and backgrounds in-the-wild human-object interactions. (2) As mentioned in a reply to the second question of the examiner, I would like to use a dataset that has the ground truth 3D appearance of the human, object and scene available, so that I can compare the perceptual scores for video-based and 3D-based human-object interaction methods.

[W5] Clarification: Page 25, Fig. 13: For clarification, the skeleton keypoints are actually annotated by human annotators, after which the parameters of the hand model are optimized to minimize the loss between the keypoints of the hand and the annotated keypoints. This piece of information in the thesis is arguably 'awkwardly' presented and could have been clarified better or alternatively entirely removed. Please refer to the DexYCB paper for more details (page 2, right side, paragraph 2, "To acquire accurate ground-truth 3D pose…"). Skeleton conditioning misnaming: You are right, I apologize for this important oversight.

Main extensions for a paper submission: (1) Use a more diverse dataset, see reply on **[W4]**, (2) Perform a noise sensitivity analysis, see reply on **[W3]** (3) Add a point-tracking evaluation method, see **[W2]**. If I had unlimited time I would focus on two fundamental challenges. (1) I hypothesize that the method struggles with consistent textures and shapes because the object in the first frame is not encoded by SVD in enough detail. I would like to train a method that allows for communication between the frozen encoder of SVD and the trainable encoder of ControlNet, in line with variant A or B in ControlNet-XS (https://vislearn.github.io/ControlNet-XS/). In this way, the ControlNet encoder could instruct the encoder of SVD to encode more detailed information on the object, hopefully allowing our method to keep textures and shapes more consistent. (2) I would like to use a 3D control signal rather than a pseudo-3D control signal to control the generation. This would require a different manner of supplying the control signal to ControlNet or SVD directly.